



Systematic literature review on sentiment analysis using transformers

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Abstract

With the rapid expansion and growing complexity of social networks, developing effective approaches for extracting meaningful knowledge from the vast volumes of user-generated data has become an essential challenge. Among these approaches, sentiment extraction and analysis— one of the core tasks in natural language processing (NLP)—plays a critical role in uncovering users' attitudes, opinions, and emotions. Insights gained from such analysis can inform strategic decision-making across a wide range of domains, including marketing, public policy, healthcare, and service management. Over the past decade, sentiment analysis has attracted increasing attention within interdisciplinary research, with many studies leveraging artificial intelligence-based techniques. In particular, transformer architectures have emerged as leading methods in NLP, primarily due to their capacity to model long-range textual dependencies and to generate richer, more precise semantic representations. This paper presents a systematic literature review of research on transformer-based sentiment analysis. Specifically, we categorize and analyze different models, widely adopted benchmark datasets, evaluation metrics, as well as the challenges and diverse applications associated with these methods. A qualitative assessment of the reviewed studies, based on established systematic review standards, highlights the progress and maturity of the field from technical, linguistic, and applied perspectives. Furthermore, we analyze current research trends and identify promising directions for future work. The findings of this review underscore both the substantial growth in research output and the increasing significance of transformer-based sentiment analysis, reflecting its dynamism, innovation, and impact on interdisciplinary studies.

Keywords Systematic literature review · SLR · Sentiment analysis · Transformers · LLMs · Quality assessment

1 Introduction

In recent years, the exponential increase in digital data alongside the widespread use of social media platforms has made the analysis of customer feedback and online reviews indispensable across various domains such as marketing, public opinion analysis, healthcare, financial markets, and social media content analysis. Since the early 2000s, sentiment analysis has emerged as one of the most dynamic research areas within natural language processing (NLP) [1]. According to Polaris Market Research [2], the global sentiment analysis market was valued at \$3.15 billion in 2021 and is projected to reach \$9.80 billion by 2030, growing at a compound annual growth rate (CAGR) of 14.4%.

Sentiment analysis, also known as opinion mining, involves the computational examination of individuals' opinions, emotions, evaluations, ratings, attitudes, and feelings towards entities such as products, services, organizations, individuals, issues, events, topics, and their attributes, with the aim of determining polarity, subjectivity, and intensity [3–5]. Traditionally, sentiment analysis relied heavily on rule-based approaches applied to simple texts, utilizing lexicons of positive and negative words. The advent of machine learning techniques marked a significant turning point, introducing algorithms such as Naive Bayes, Support Vector Machines, and Decision Trees to enable sentiment classification of more complex textual data [6].

The rise of social media platforms like Twitter and Facebook has generated vast amounts of short and informal text data [7]. Subsequently, more sophisticated deep learning models have been developed, such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Convolutional Neural Networks

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(CNN) [8]. These models have demonstrated strong performance in complex sentiment analysis tasks owing to their ability to understand context and model nonlinear relationships [9].

Transformers represent a state-of-the-art neural network architecture designed explicitly for textual data processing. They have gained widespread adoption in sentiment analysis due to their superior capability to model long-range dependencies and extract precise semantic representations from text [10]. Transformers can detect intricate relationships among words and phrases within sentences, thereby enabling more accurate recognition of users' attitudes, feelings, and emotions compared to traditional deep learning models [11, 12]. The Bidirectional Encoder Representations from Transformers (BERT) model, utilizing a bidirectional learning mechanism that simultaneously considers preceding and succeeding words in a text, was developed to provide a deeper understanding of natural language [13]. Unlike prior models that processed text unidirectionally, BERT analyzes the entire sequence at once. RoBERTa, an optimized variant of BERT with enhancements such as larger training datasets, removal of certain pretraining constraints, and other refinements, has demonstrated superior performance relative to BERT [14]. Fine-tuning these models has enabled them to achieve remarkable accuracy across diverse tasks, including text classification, question answering, and sentiment analysis.

In recent years, large language models (LLMs) have emerged as powerful tools in sentiment analysis. GPT-3, notable for its vast scale and versatile text generation and task-solving capabilities, has garnered significant attention [15]. These models excel in few-shot or zero-shot sentiment classification scenarios, possess the ability to comprehend complex contexts and linguistic nuances, and can perform sentiment analysis across multimodal data, including text, audio, and images.

The diversity of transformer-based methods and architectures—ranging from complex hybrid designs to varied application requirements such as computational resource availability, domain specificity, domain adaptation needs, data scarcity, or limited datasets—implies that no single approach suffices for all sentiment analysis scenarios. This makes it challenging to align the capabilities of each model or method with the specific use cases of different contexts. Accordingly, researchers employ various strategies, which we categorize as context-adaptive models, large language models, lightweight models, and hybrid architectures.

Context-adaptive models are trained on diverse domains to enhance their adaptability across varying contexts before fine-tuning for sentiment analysis in specific fields [16]. They also include the models that are developed from the ground up with domain-specific architectures and configurations tailored to particular datasets (e.g., financial, social, or

medical data) [17]. The primary objective is to customize the model to the unique characteristics and requirements of each domain, thereby enhancing accuracy and efficiency. Large language models are transformer-based architectures with an extensive number of parameters, capable of understanding and processing language at scale [18, 19]. Utilizing techniques like in-context learning, sparse learning, and prompt engineering, these models perform sentiment analysis at more granular and sophisticated levels. Their flexibility and generalizability allow application across multiple domains and languages. Lightweight models are designed or re-engineered to reduce computational complexity and optimize performance for resource-constrained environments [20, 21]. Techniques such as model compression, pruning, and knowledge distillation are employed to minimize model size and computational demands without significantly compromising sentiment analysis performance. These models are particularly suitable for applications requiring real-time processing or operating under hardware limitations. Many studies have also explored hybrid architectures combining transformers with other deep learning approaches, statistical methods, or linguistic and symbolic knowledge sources (e.g., semantic lexicons, knowledge bases, or ontologies) [22, 23]. This integration aims to leverage the distributed learning strengths of transformers alongside structural or conceptual knowledge from other sources to improve sentiment analysis outcomes.

Despite notable advances, several challenges persist due to the multidimensional nature of data and the complex characteristics of users. Linguistic complexity remains a significant issue; words may carry different meanings depending on context, and emotions are inherently subjective, varying across individuals, cultures, and situations. Multilingual or cross-lingual sentiment analysis introduces additional linguistic difficulties. Many existing methods inadequately address these complexities—for instance, several studies relying predominantly on style-based models have overlooked finer content nuances [20, 21]. To bridge this gap, we classify relevant studies into those that minimally consider linguistic intricacies, mainly employing lightweight models, and those that explicitly account for them, including approaches leveraging large language models.

Given the breadth and complexity of research on transformer-based sentiment analysis, there is an urgent need for a comprehensive, systematic review to categorize and assess existing methods, as well as to identify research challenges and gaps. A systematic literature review (SLR) offers a targeted investigation of a specific research landscape, evaluating various facets of the topic and comparing related studies. Moreover, it analyzes multiple dimensions of the field to aid researchers and highlight future research directions. These types of research have emerged in recent years, e.g., Ziarani and Ravanmehr [24], Seyfossadat and Ravanmehr [25] conducted systematic reviews in various AI fields

to provide a structured and methodical development framework for the research community. In our study, recognizing the significance of transformer-based sentiment analysis and the absence of a thorough and appropriate systematic review in this area, we propose an SLR dedicated to this field. Our approach begins with an examination of different types of sentiment analysis and associated challenges, followed by defining five research questions to guide the review process. Based on this framework, 347 articles published between Jan 2020 and August 2025 were examined. Subsequently, various transformer-based sentiment analysis approaches were categorized, reviewed, and analyzed, focusing on aspects such as techniques, key strengths and limitations, datasets and methodologies, and evaluation criteria. Finally, the quality of the studies was assessed using multiple parameters to evaluate the rigor of research conducted in this domain and to identify emerging trends. The study concludes by outlining future research opportunities informed by existing gaps.

1.1 Related surveys

Although numerous review studies have been published in the field of sentiment analysis, this work addresses a critical gap in the literature by conducting a systematic literature review (SLR) focused specifically on transformer-based models during the contemporary period (Jan 2020–Aug 2025).

Prior reviews can generally be divided into two categories: broad-scope surveys and domain-specific investigations. Some reviews, such as [26], offer a comprehensive and historical overview, tracing the evolution of sentiment analysis from traditional methods to modern approaches. Others, like [27], adopt an interdisciplinary lens by simultaneously examining sentiment recognition, affective computing, and sentiment analysis across multiple modalities, including physiological signals. In contrast, the present study narrows its scope to a single technology (transformers) and a single domain (sentiment analysis), thereby providing a more in-depth and fine-grained exploration of recent architectures and applications. For instance, while [28] surveys sentiment analysis in social networks with attention to temporal, causal, and industrial dimensions—yet devotes only limited discussion to transformer-related challenges—this review distinguishes itself by concentrating exclusively on transformer architectures and by systematically examining associated models, datasets, and evaluation metrics. Similarly, whereas [29] presents an extensive overview of all deep learning paradigms before 2023, our study delivers a more specialized analysis by restricting its scope to transformer-based models within the 2020–2025 timeframe, complemented by a structured qualitative evaluation. [30] focuses exclusively on aspect-based sentiment analysis (ABSA), while [31] examines

sentiment analysis in the context of online learning environments (MOOCs). Likewise, [32] addresses the challenge of low-resource languages, and [33] explores multimodal approaches combining text and images. In contrast to these vertical examinations, the current review adopts a horizontal perspective, analyzing transformer applications across all subfields and domains in an integrated fashion.

Another line of work focuses on surveys that restrict their scope to transformer-based technologies within the domain of sentiment analysis. For example, Bashiri et al. [34] adopt a largely quantitative perspective by comparing the performance of multiple transformer models, including BERT, RoBERTa, XLNet, ELECTRA, DistilBERT, ALBERT, T5, and GPT, across diverse datasets, thereby highlighting their respective strengths and weaknesses. Gautam et al. [35] present a survey on the role of pretrained language models (PLMs) and large language models (LLMs), including BERT, RoBERTa, GPT-4, and LLaMA-2, in sentiment classification. The study compares these models with lexicon-based and traditional deep learning approaches, examines fine-tuning and domain adaptation strategies, and discusses key limitations such as computational cost, bias, and ethical considerations. In contrast to [34] and [35], which focus on the impact and evolution of pretrained language models (PLMs) and large language models (LLMs) in sentiment classification while comparing them with earlier approaches, our SLR presents a systematic review of the most recent developments in this scope and introduces a novel technical taxonomy that categorizes transformer models into context-adaptive, large language model, lightweight, and hybrid variants, thereby providing a structured perspective that is absent from prior transformer-focused reviews.

In summary, the main contributions and distinguishing features of our review are threefold. First, it frames its analysis within the transformer era (Jan 2020–Aug 2025), capturing the most recent developments. Second, rather than relying on broad categories, it introduces a technical taxonomy of transformer architectures (adaptive, hybrid, lightweight, and LLM-based models). Third, it presents the first SLR, which not only synthesizes prior work but also incorporates diverse analyses, such as a comprehensive qualitative assessment (QAQ) of the literature in this domain.

1.2 Contributions

This paper conducts a comprehensive systematic literature review (SLR) on transformer-based approaches to sentiment analysis. It provides an in-depth examination of the methods, models, datasets, and applications employed across diverse domains. The primary goal is to provide researchers with a structured roadmap that both guides future investigations and offers insights for developing sentiment analysis techniques

capable of overcoming technical limitations while enhancing accuracy and efficiency.

The main *contributions* of this review are as follows:

- Identification and classification of sentiment analysis tasks, associated challenges, and real-world applications,
- A detailed comparative assessment of the key datasets commonly used in sentiment analysis,
- A review and evaluation of transformer-based sentiment analysis methods, with emphasis on their advancements, innovations, and limitations,
- Analysis of the core evaluation metrics adopted in sentiment analysis, along with an assessment of their adequacy and comprehensiveness,
- A qualitative assessment of the published literature, following established standards for systematic reviews,
- Development of practical guidelines—covering models, datasets, evaluation methods, and metrics—derived from the synthesis, aimed at supporting the design of innovative transformer-based approaches across different sentiment analysis settings,
- Identification of research gaps and unresolved challenges, coupled with a discussion of promising directions for future work in transformer-driven sentiment analysis.

1.3 Paper organization

The structure of the paper is organized as follows. Section 2 provides an overview of different approaches to sentiment analysis, discusses the associated challenges, and outlines its practical applications. Section 3 details the methodology adopted for conducting the systematic literature review. Section 4 reports the review findings and addresses the research questions. Section 5 highlights the identified research gaps and explores potential directions for future work. Finally, Sect. 6 concludes the paper with a summary of key insights.

2 Basic concepts

To establish a comprehensive foundation for this review, this section outlines the landscape of sentiment analysis in the context of transformer architectures. We first categorize the primary analysis tasks and highlight the specific applicability of transformer-based models to each. Subsequently, we identify persistent technical and linguistic challenges inherent to the field. Finally, we present a structured analysis of diverse application domains, detailing the specific obstacles and tailored transformer solutions relevant to each sector.

2.1 Types of sentiment analysis

There are various tasks related to sentiment analysis, which differ depending on the research objective, the application domain, and the level of granularity required. Based on the literature reviewed, we categorize these tasks below and highlight the applicability of transformer-based models in each domain:

2.1.1 Polarity-based sentiment analysis

This approach determines whether a sentiment is positive, negative, or neutral. More advanced versions introduce graded categories, such as “very positive” or “very negative.” Polarity-based analysis is widely used in contexts such as customer reviews, social media monitoring, and product feedback assessment. Transformer architectures are particularly well-suited for this task because their self-attention mechanism effectively captures context-dependent shifts in polarity, such as negation (e.g., “not bad”) or sarcasm, which traditional bag-of-words models often miss. In this context, Souza et al. [16] demonstrated that fine-tuning BERT models significantly outperforms traditional methods in binary sentiment classification tasks for Portuguese texts.

2.1.2 Emotion-based sentiment analysis

Unlike polarity-based methods, this approach detects specific emotions—such as joy, anger, sadness, fear, or surprise—rather than general positivity or negativity. It is particularly valuable for brands and services (e.g., mental health care) that need to respond empathetically. Transformers are highly effective in this domain because their deep contextual embeddings can detect subtle semantic triggers and implicit emotional cues that go beyond simple keyword matching [36, 37]. For instance, Lin et al. [38] applied BERT-based models to successfully classify fine-grained emotions (e.g., fear, anger) in COVID-19 related tweets, while Chandra et al. [39] utilized them for multi-label emotion categorization in news articles.

2.1.3 Aspect-based sentiment analysis (ABSA)

Instead of assessing sentiment at the document or sentence level, ABSA evaluates opinions related to specific attributes or features of an entity—for instance, the battery life or camera quality of a smartphone. The primary reason for the success of transformers in ABSA is their ability to model long-range dependencies; the attention mechanism can accurately link a specific “aspect” word to its corresponding “sentiment” word, even if they are separated by a long distance in the sentence. Our review indicates that ABSA is one of the most active research areas, particularly utilizing hybrid

architectures. Models such as HybBERT [40] and Prompt-CSR [23] successfully leverage transformers to capture these complex dependencies between aspects and sentiment polarities.

2.1.4 Fine-grained sentiment analysis

Fine-grained Sentiment Analysis provides a more nuanced categorization of sentiment, extending beyond simple polarity to include graded categories (e.g., “very positive” to “very negative”) or numerical rating scales. Transformers are applicable here because their dense vector representations can capture the intensity and degree of sentiment modifiers (e.g., distinguishing “good” from “extremely good”) through multi-class classification or regression heads. For instance, studies such as Wang et al. [41] and Hou et al. [42], utilizing the CMU-MOSI and CMU-MOSEI datasets, employ metrics like Acc-7 to evaluate sentiment on a fine-grained 7-point scale (ranging from -3 to $+3$).

2.1.5 Intent-based sentiment analysis

This type goes beyond emotional polarity to infer the underlying intent behind the text—such as filing a complaint, requesting assistance, or offering a suggestion. Large Language Models (LLMs) and transformers are superior in this task due to their extensive pre-training on diverse instructions, which allows them to understand pragmatic context and infer user goals that are not explicitly stated. Recent studies, such as Benayas et al. [43], highlight the superior capability of LLaMA2 and RoBERTa compared to traditional methods in distinguishing between diverse user intents with high accuracy.

2.1.6 Subjectivity/objectivity analysis

This method classifies text into subjective statements (opinion or feeling-based) and objective statements (fact-based). It is often used as a preprocessing step to filter out neutral information. Transformers aid this process by distinguishing between descriptive language (objective) and evaluative language (subjective) through the analysis of syntactic structures and tonal patterns. In our review, studies involving “opinion mining,” such as Scerri et al. [44], inherently rely on this distinction. Furthermore, widely used multimodal datasets like CMU-MOSI include specific annotations for subjectivity to aid in training transformer models.

2.1.7 Multimodal sentiment analysis

Multimodal approaches analyze sentiment across different data types—such as text, audio, and images. This is particularly useful for video reviews and social media content where

tone and facial expressions convey emotion. Transformers have revolutionized this field through “Cross-Modal Attention,” a mechanism that aligns and integrates heterogeneous data streams, allowing the model to resolve ambiguities (e.g., detecting irony when a positive text is spoken with a negative tone). Frameworks proposed by Ai et al. [45] and Wang et al. [41] demonstrate that transformers can effectively align and integrate these diverse data streams.

2.1.8 Success and challenges in transformer-based tasks

While transformers have been applied across all these tasks, our systematic review suggests they have been particularly dominant and successful in ABSA and Multimodal tasks, where capturing long-range dependencies and cross-modal interactions is crucial. However, significant challenges persist across these domains. First, Domain Generalization remains a critical hurdle; models pre-trained on generic corpora often suffer from performance degradation when applied to specialized fields (e.g., medical or financial texts) without extensive fine-tuning. Second, Data Imbalance and Scarcity pose severe limitations, particularly for Emotion-based and Fine-grained analysis, where the lack of labeled samples for rare emotions or extreme polarity intensities leads to biased predictions. Third, linguistic nuances such as Sarcasm and Irony continue to baffle even sophisticated models like BERT, as detecting them requires deep pragmatic understanding beyond surface-level semantics. Additionally, applying these models to Noisy/Informal Language (e.g., social media slang) requires specialized adaptation, as seen in TM-BERT [46]. Finally, the Computational Cost of deploying these heavy architectures remains a substantial barrier, necessitating further research into lightweight variants like DistilBERT for resource-constrained environments.

2.2 The challenges of sentiment analysis

Despite substantial progress in deep learning and transformer-based models, sentiment analysis continues to face several persistent challenges that limit model performance across diverse scenarios. The most prominent challenges include:

2.2.1 Linguistic, cultural, and structural diversity

Texts vary widely in terms of language, dialect, syntax, and cultural context. A word with a positive connotation in one culture may carry a negative or neutral meaning in another. This issue is particularly pronounced in low-resource languages or those with complex morphological structures, where the accuracy of general-purpose models—often developed primarily for English—tends to decline [16, 47–49].

2.2.2 Ambiguity, sarcasm, and layered emotions

Detecting implicit sentiment, irony, or sarcasm remains a major challenge. Many current models misinterpret texts that express mixed or nuanced emotions (e.g., “I liked it, but I expected more”). Accurately capturing such sentiments requires a deeper grasp of context and semantic relationships [50–53].

2.2.3 Data imbalance

Class imbalance is a common issue in sentiment datasets, where one category (e.g., neutral) often dominates others. This imbalance biases models toward the majority class, thereby reducing accuracy when classifying minority sentiments [19, 23, 54–56].

2.2.4 Noisy and informal language

Social media platforms, such as Twitter, present additional challenges due to the frequent use of slang, emojis, abbreviations, hashtags, and misspellings. Properly handling such content demands robust preprocessing techniques and models capable of interpreting highly informal language [57–59].

2.2.5 Multimodal data

When sentiment is expressed through multiple channels—such as text combined with audio, images, or video—effective analysis requires models that can seamlessly integrate information across modalities [60–63].

2.2.6 Computational cost

Although large transformer models such as BERT achieve high accuracy, their heavy computational requirements limit their practical deployment. To enable real-world applications, it is essential to develop lighter models that reduce computational costs without sacrificing performance [20, 21, 64, 65].

2.3 Application domains: challenges and transformer-based solutions

Sentiment analysis, as a significant branch of natural language processing (NLP), has transcended general text classification to address complex needs across diverse sectors. A systematic review of recent studies reveals that transformer-based models have been widely adopted across a broad spectrum of industries. As illustrated in Fig. 1, social media platforms dominate the research landscape. However, our systematic review also reveals a rapidly growing application

in specialized fields such as cybersecurity, smart tourism, and decentralized finance.

2.3.1 Social media and public opinion

Social media platforms (Twitter/X, Reddit, Weibo, Facebook) represent the most extensive application domain. Research here goes beyond simple brand monitoring to encompass complex sociological and political analyses. Studies have utilized transformers to analyze political discourse, such as public attitudes toward nuclear energy policies [66]. Furthermore, crisis response analysis—ranging from the Jakarta flood [67] to the Russia-Ukraine war discourse on YouTube [68]—demonstrates the utility of these models in understanding real-time public sentiment.

- **Challenges:** The primary obstacle is the prevalence of unstructured and “noisy” text, characterized by slang, emojis, hashtags, and code-mixing (e.g., Hinglish [69] or code-mixed Bengali [70]). Additionally, detecting subtle linguistic nuances like sarcasm remains computationally difficult.
- **Transformer Solutions:** Researchers have addressed these issues by developing domain-adaptive models. For instance, Riaz et al. [46] introduced TM-BERT, optimized specifically for noisy tweets. Other approaches leverage multilingual transformers to handle code-mixed data effectively in diverse linguistic regions [71].
- **Datasets:** Research in this domain extensively utilizes large-scale benchmarks such as Sentiment140 for general sentiment classification and Twitter US Airline for customer feedback. Additionally, specialized corpora like the sentiment@USNavy dataset are used to analyze directed social media interactions in specific organizational contexts.

2.3.2 Marketing, E-commerce, and tourism

This domain is critical for business intelligence and encompasses user-generated reviews of products (electronics, books, fashion) and services. A significant portion of research focuses on the hospitality sector, analyzing reviews of hotels and restaurants to assess service quality [72]. Emerging niche applications also include analyzing customer feedback for bakeries [73].

- **Challenges:** Users often express mixed sentiments about different attributes in a single review (e.g., “The location was great, but the room was dirty”). Standard polarity analysis fails to capture this granularity.

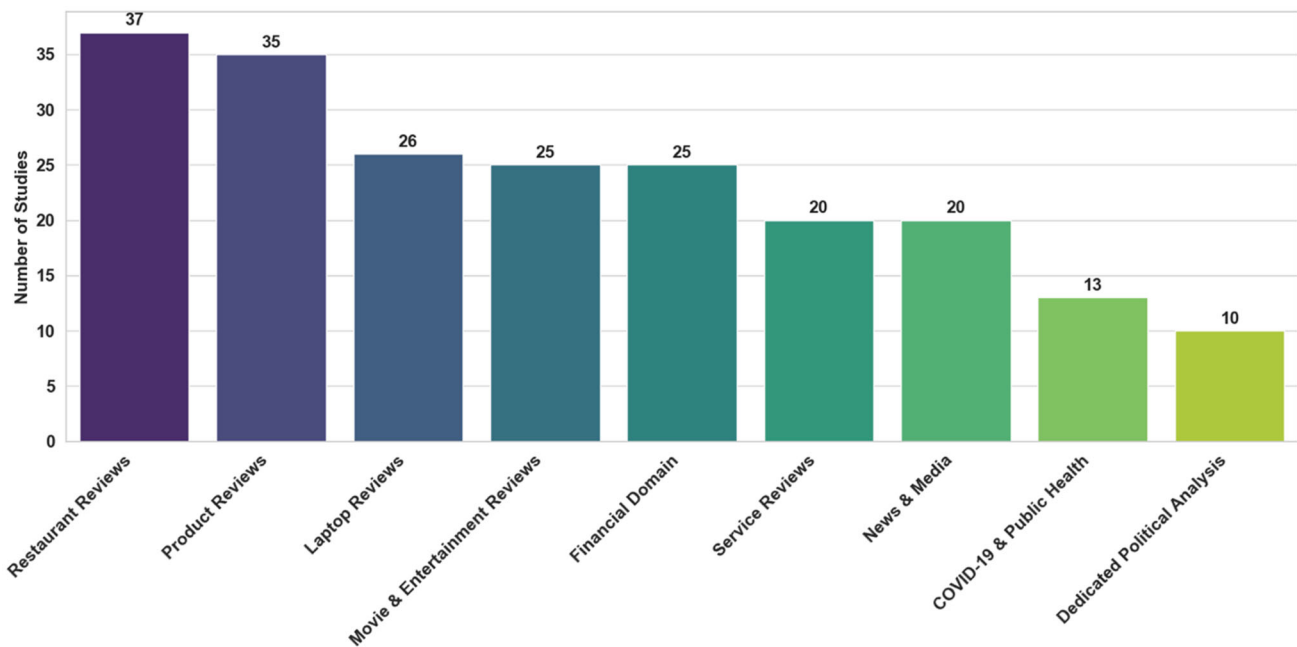


Fig. 1 Application domains of sentiment analysis using transformers

- **Transformer Solutions:** This domain relies heavily on Aspect-Based Sentiment Analysis (ABSA). Hybrid models, such as HybBERT [40], are extensively used to isolate specific aspects (e.g., "room quality" vs. "service") and classify sentiment for each, providing actionable insights for service improvement.
- **Datasets:** The evaluation of models in this sector heavily relies on the IMDB dataset and Rotten Tomatoes for movie reviews. In the e-commerce and hospitality sectors, the Yelp dataset and Amazon Musical Instruments reviews serve as primary benchmarks, alongside the SemEval task datasets (e.g., Restaurant and Laptop reviews).

2.3.3 Financial markets and fintech

Sentiment analysis in finance has evolved from simple stock prediction to complex applications in the fintech ecosystem. Beyond analyzing financial news and stock reports to predict market volatility [74], recent studies explore Decentralized Finance (DeFi). This includes detecting scam tokens and fraud by analyzing community sentiment alongside transaction logs [75].

- **Challenges:** Financial texts contain highly specialized jargon (e.g., "bearish," "bullish," "liability") where general sentiment lexicons fail. Moreover, the need for real-time processing to detect fraud adds a layer of complexity.

- **Transformer Solutions:** General models often underperform; thus, domain-specific models like FinBERT are preferred. Singh et al. [74] demonstrated a hybrid framework using FinBERT combined with Bi-LSTMs to effectively correlate news sentiment with stock indices. For crypto scams, transformer-based frameworks like DeFiTrust [75] are employed to analyze social sentiment patterns.
- **Datasets:** Studies typically employ the FiQA dataset for financial question-answering and sentiment tasks.

2.3.4 Healthcare and crisis management

The application of sentiment analysis in healthcare has surged, particularly for public health monitoring. Studies track public attitudes toward vaccines (e.g., Pfizer/COVID-19) [76] and even aesthetic enhancements trends on visual platforms like Instagram [77].

- **Challenges:** Medical texts require high precision due to the critical nature of the domain. Key issues include interpreting medical terminology accurately and managing misinformation in public discourse.
- **Transformer Solutions:** Models pre-trained on scientific or biomedical corpora have shown superior performance. For instance, Lin et al. [38] utilized CT-BERT to capture the specific context of health-related tweets. Integrating advanced architectures has further refined crisis monitoring; Agrawal et al. [78] demonstrated that a hybrid

framework using DistilBERT effectively balances computational efficiency with accuracy in tracking vaccination sentiment. Similarly, Kumar et al. [79] proposed an ensemble of BERT and CNN to robustly capture both global context and local features in COVID-19 discourse, outperforming single-model baselines.

- **Datasets:** The primary data sources in this domain are pandemic-related collections, such as the CV-SAT dataset, which focuses on COVID-19 vaccination sentiment, and the SenWave dataset used for multi-label emotion classification in social media texts during health crises.

2.3.5 Politics, governance, and sustainability

This emerging domain leverages sentiment analysis to enhance civic engagement and policy-making. Applications include analyzing citizen feedback on urban planning initiatives, such as transport redesigns in Malta [44], and evaluating public digital resources for sustainable development [80].

- **Challenges:** Capturing the "stance" and subtle political biases in formal government complaints versus informal protests is challenging.
- **Transformer Solutions:** Topic-aware transformers (e.g., BERTopic integrations) are used to identify key societal concerns from unstructured citizen feedback [81], enabling governments to prioritize policy responses.
- **Datasets:** Political sentiment analysis often relies on news media corpora, such as The Guardian articles dataset, to track public sentiment and multi-label emotions over time. Additionally, Central Bank Speeches are analyzed to understand policy-related sentiment dynamics.

2.3.6 Cybersecurity and software engineering

A niche but growing area involves applying sentiment analysis to technical environments. This includes detecting intrusion attempts via sentiment forensics [82].

- **Challenges:** Texts in this domain often contain technical jargon mixed with natural language. Standard models struggle to interpret the semantic meaning of such specialized content.
- **Transformer Solutions:** Transformers are adapted to understand the context of technical discussions, helping to distinguish between technical anomalies and normal interactions [82].
- **Datasets:** In technical domains, datasets like Toxic Comment Classification are adapted to detect harmful, toxic, or anomalous content within digital interactions, serving

as a proxy for identifying security threats or community health issues in software platforms.

2.3.7 Multimodal applications and education

The field is increasingly shifting towards multimodal analysis, processing text alongside audio and video. This includes analyzing video reviews and memes (e.g., Hinglish memes [69]). In education, sentiment analysis is used to assess perceptions of AI tools like ChatGPT in learning environments [81] and analyze feedback in virtual learning environments [83].

- **Challenges:** The primary challenge is the alignment of heterogeneous data streams—sentiment is often conveyed through tone (audio) or facial expressions (visual), which text-only models miss.
- **Transformer Solutions:** Multimodal transformers, such as the frameworks proposed by Ai et al. [45] and Wang et al. [41], fuse textual embeddings with acoustic and visual features to capture the full sentiment spectrum effectively.
- **Datasets:** Multimodal research is strongly anchored in the CMU-MOSI and CMU-MOSEI datasets, which provide aligned video, audio, and text data for sentiment intensity analysis. These benchmarks are standard for evaluating fusion strategies in transformer-based models.

Table 1 provides a summary of application domains, primary challenges, transformer-based solutions, and common dataset types in the reviewed studies.

3 Methodology

In the field of computer science, systematic literature reviews (SLRs) have gained growing significance. The primary objective of an SLR is to systematically and rigorously identify, analyze, and synthesize relevant studies to address specific research questions. In this study, the SLR was conducted in accordance with the established guidelines [84]. An SLR typically begins with the formulation of research questions, which are introduced in Sect. 3.1. Section 3.2 then outlines the extended review process undertaken to address these questions. Finally, Sect. 3.3 discusses the article selection procedure, which was carried out following the PRISMA framework, and specifies the inclusion and exclusion criteria applied.

Table 1 Summary of application domains

Application domain	Primary challenges	Transformer-based solutions	Common datasets
Social Media & Public Opinion	Noise, slang, sarcasm, code-mixing	Domain-adaptive models (TM-BERT), Multilingual transformers	Sentiment140, Twitter US Airline, Toxic Comment Classification
Marketing, E-commerce & Tourism	Mixed sentiments, aspect extraction	Aspect-Based Sentiment Analysis (ABSA), Hybrid models	IMDB, Yelp, Amazon Musical Instruments, Rotten Tomatoes
Financial Markets	Specialized jargon, market volatility correlation	Domain-specific pre-training (FinBERT), Hybrid LSTM frameworks	FiQA, Financial News Headlines, Central Bank Speeches
Healthcare & Crisis Management	Medical terminology, misinformation	Bio-medical pre-training (BioBERT, CT-BERT), Ensembles	CV-SAT, SenWave, COVID-19 Twitter Datasets
Politics & Governance	Stance detection, subtle bias	Topic-aware models (BERTopic)	The Guardian Articles, Central Bank Speeches, sentiment@USNavy
Cybersecurity & Software Engineering	Technical jargon, anomaly context	Context-aware transformers	Toxic Comment Classification
Multimodal & Education	Aligning heterogeneous data	Multimodal Fusion Transformers	CMU-MOSI, CMU-MOSEI

Table 2 Research questions

Number	Research question
RQ1	How has sentiment analysis evolved with the advent of transformers?
RQ2	Which transformer-based models have been employed for sentiment analysis?
RQ3	What are the key datasets used for evaluating transformer-based sentiment analysis models?
RQ4	Which metrics are used to evaluate transformer-based sentiment analysis models?
RQ5	How has the maturity and scientific quality of transformer-based sentiment analysis research evolved in recent years?

Table 3 Research search strings

Search string	Related RQ(s)
("Sentiment Analysis" OR "Opinion Mining") AND ("Transformer") AND ("Evolution" OR "Improvement" OR "Progress") AND ("Meaning" OR "Concept" or "Definition")	RQ1
("Sentiment Analysis") AND ("Transformer-based models" OR "Pre-trained Language Models" OR "LLM")	RQ2, RQ5
("Sentiment Analysis") AND ("Method" OR "Technique" OR "Approach" OR "2.3.7")	RQ2
("Sentiment Analysis") AND ("Transformer") AND ("Dataset" OR "Benchmark" OR "Corpus")	RQ3
("Sentiment Analysis") AND ("Transformer") AND ("Measure" OR "Evaluation" OR "Measurement" OR "Evaluating" OR "Metric")	RQ4

3.1 Research questions

From the preliminary review, four initial research questions (Q1–Q4) were formulated. Subsequently, an in-depth examination of various studies in the field of sentiment analysis led to the development of the primary research question (Q5). For each research question (Table 2), corresponding search fields were defined (Table 3) to facilitate a structured analysis.

Sentiment analysis remains one of the most challenging tasks in natural language processing, as the interpretation of sentiment in a sentence can vary significantly depending on context, wording, and syntactic structure. In recent

years, the introduction of the transformer architecture has marked a turning point in this domain. Accordingly, the first research question (RQ1) investigates how sentiment analysis has evolved with the advent of transformer-based models.

With the expansion of transformer models, increasingly sophisticated variants have been developed for sentiment analysis. This leads to the second research question (RQ2): Which transformer-based models have been employed for sentiment analysis?

Evaluating these models requires access to standardized and reliable datasets. Therefore, the third research question (RQ3) focuses on identifying the key datasets used to assess transformer models in sentiment analysis: What are the key datasets used for evaluating transformer-based sentiment analysis models?

Moreover, assessing the performance of different models necessitates the use of well-defined and trustworthy evaluation metrics. For this reason, the fourth research question (RQ4) addresses the following: Which metrics are used to evaluate transformer-based sentiment analysis models?

Finally, with the emergence of novel approaches such as large language models (LLMs), sentiment analysis has continued to evolve. Thus, the fifth research question (RQ5) examines the following: How has the maturity and scientific quality of transformer-based sentiment analysis research evolved in recent years?

3.2 SLR process

In this study, the systematic literature review (SLR) process was carried out in five stages:

Step 1. Formulating the research questions (Table 2), identifying relevant keywords, and defining the scope of inquiry.

Step 2. Specifying the research domains to be searched. Articles were retrieved from reputable digital libraries, including Elsevier, Springer, Wiley, IEEE, ACM, IOS Press, Taylor & Francis, MDPI, and Emerald Insight.

Step 3. Conducting an initial screening of the articles to verify their relevance to transformer-based sentiment analysis models. At this stage, the abstract, introduction, and conclusion of each article were examined.

Step 4. Performing a broader assessment of the selected articles beyond the abstract, introduction, and conclusion. The full texts were reviewed at a high level to determine whether they addressed the research questions. In this study, an article was considered relevant if it provided at least one answer to any of the research questions.

Step 5. Conducting an in-depth review of the eligible articles and extracting five key elements:

- Sentiment analysis method
- Research datasets
- Evaluation metrics for sentiment analysis
- Application domain
- Research gaps

Figure 2 depicts the process of the SLR.

3.3 Study selection process

In this study, we adhered to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines to ensure that the most relevant articles were selected with respect to the scope and depth of the research. PRISMA provides a structured framework that reflects methodological advancements in the identification, screening, evaluation, and synthesis of studies. The guidelines were initially introduced by Moher et al. [85], and an updated version, known as PRISMA 2020, was published in 2021 [86].

The PRISMA framework consists of four main stages:

- **Stage 1, Identification:** Focuses on key considerations when retrieving articles from relevant sources, including the credibility of the source.
- **Stage 2, Screening:** Applies criteria for the preliminary filtering of articles, typically based on the title and abstract.
- **Stage 3, Eligibility:** Involves reviewing the full text of articles to determine their relevance to the research.
- **Stage 4, Inclusion:** Presents the final set of articles selected at the end of the process.

The outcomes of article selection at each stage of the PRISMA process are illustrated in Fig. 3, which summarizes the number of studies assessed across the different phases.

To ensure the selection of high-quality and relevant studies, specific inclusion (IC) and exclusion (EC) criteria were defined. As detailed in Table 4, these criteria guided the screening process to filter articles based on publication date, language, topic relevance, and document type.

4 Results of systematic literature review

In this section, we summarize the results and responses to the research questions (Table 2). The relevant studies were identified and collected using predefined keywords and search strings (Table 3). Figure 4 illustrates the annual distribution of the collected articles between Jan 2020 and Aug 2025, while Fig. 5 presents the frequency of publications across different publishers. Altogether, these 347 articles—retrieved through the PRISMA procedure—form the foundation of our systematic literature review.

4.1 The impact of transformers in sentiment analysis

In response to the first research question (RQ1), “*How has sentiment analysis evolved with the advent of transformers?*”, it is essential to first clarify the concept of sentiment analysis as a key subfield of natural language processing (NLP). Broadly defined, sentiment analysis refers to the task of identifying and classifying emotional states expressed in

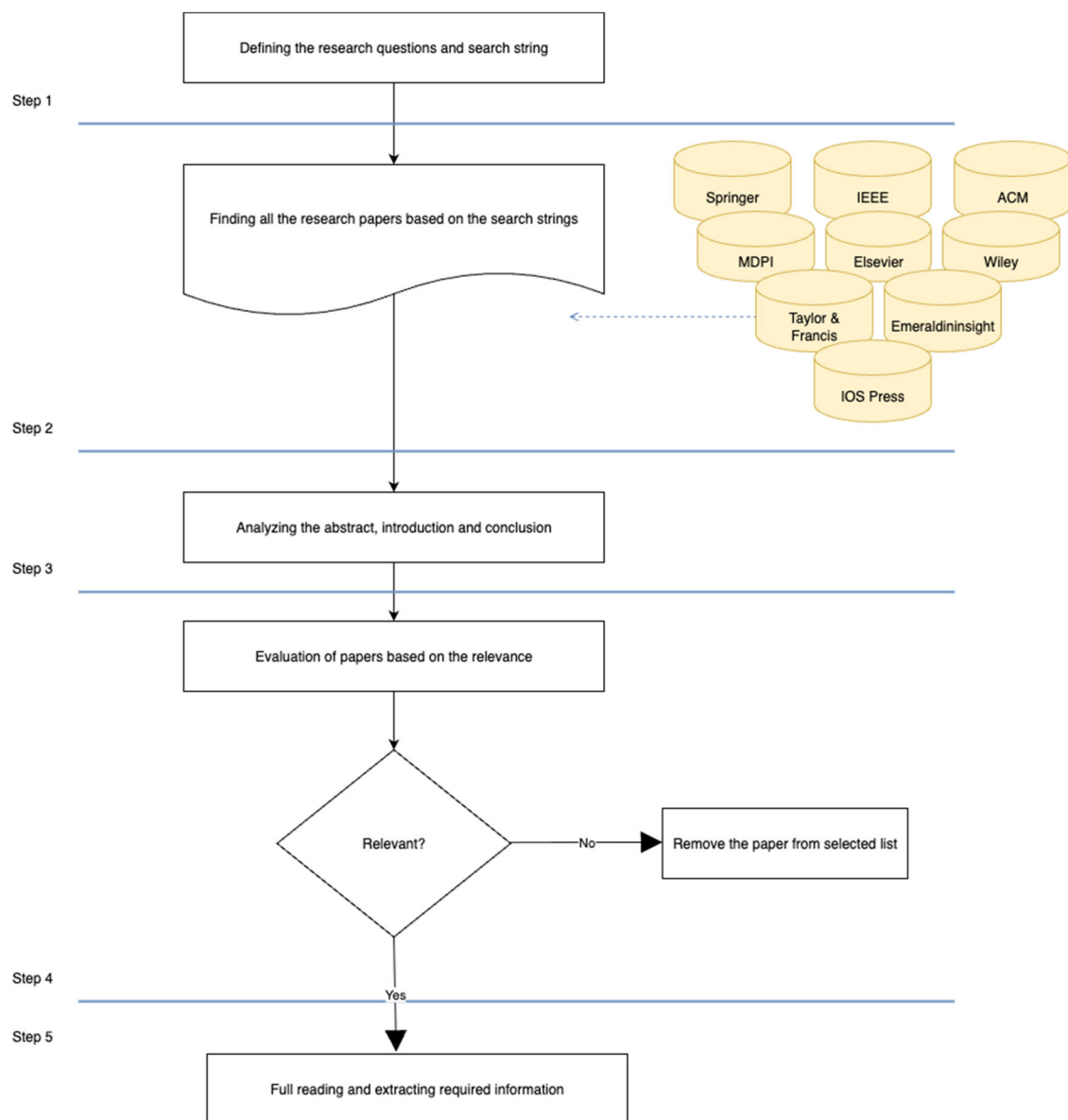


Fig. 2 The process of the SLR in this research

written text. This capability enables organizations to better interpret user feedback, make data-driven decisions, and refine their strategies to enhance user experience and customer satisfaction.

Early approaches relied on rule-based methods, often supported by sentiment lexicons. With the emergence of machine learning algorithms such as Support Vector Machines (SVM) and Naïve Bayes, researchers were able to handle more complex textual inputs [6]. The subsequent development of deep learning architectures, including recurrent neural networks (RNNs) and long short-term memory (LSTM) models, addressed some of the limitations of traditional machine

learning, such as the heavy reliance on manual feature engineering [9].

A decisive shift occurred with the introduction of transformer architectures, most notably BERT [13], RoBERTa [14], and later GPT [15]. These models largely displaced prior approaches based on manual features, bag-of-words representations, and even RNN/LSTM models. The self-attention mechanism [10] enabled transformers to capture long-range dependencies and linguistic nuances, such as sarcasm, negation, emotional intensity, and aspect-specific sentiment, far more effectively than sequence-based predecessors.

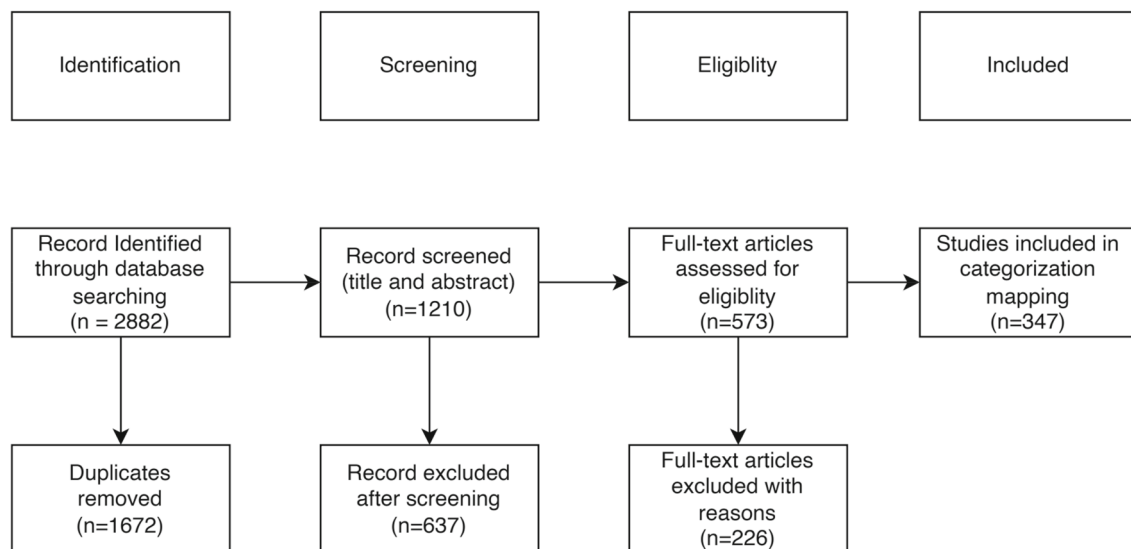


Fig. 3 PRISMA guideline for study selection in different phases

Table 4 Inclusion and exclusion criteria

Exclusion criteria (EC)	ID	Inclusion criteria (IC)	ID
Studies published before 2020	EC1	Studies published between Jan. 2020 and Aug. 2025	IC1
Non-English articles	EC2	Articles written in the English language	IC2
Studies focusing solely on traditional ML (SVM, Naïve Bayes) or non-transformer DL (RNN, CNN) without transformers	EC3	Studies specifically utilizing Transformer-based models (e.g., BERT, RoBERTa, LLMs) for sentiment analysis	IC3
Short papers, posters, abstracts, doctoral theses, or grey literature	EC4	Articles published in peer-reviewed journals or reputable conferences	IC4
Duplicate studies or articles unrelated to the specific research questions	EC5	Studies that provide explicit answers to at least one Research Question (RQ)	IC5

Another critical innovation was large-scale pretraining on massive general-domain corpora, followed by lightweight fine-tuning for domain-specific tasks. This made it feasible to achieve strong performance even with limited labeled data [87]. At the evaluation stage, metrics also evolved, moving beyond simple binary accuracy toward multi-label and multi-class classification, continuous sentiment intensity scoring [88], and aspect-based sentiment analysis [89]. Collectively, transformers have significantly advanced the accuracy, adaptability, and scalability of sentiment analysis.

In recent years, sentiment analysis has been applied across diverse domains, including marketing and customer experience, financial markets, social media and public opinion, film and entertainment reviews, health and societal crises, journalism and news coverage, as well as multimodal and low-resource language analysis.

Transformer-based sentiment analysis has also given rise to multiple methodological approaches. Adaptive models are either pretrained on diverse domains and later fine-tuned for

a specific application, or designed from scratch for domain-specific data such as financial, medical, or social contexts. Large language models (LLMs) represent another category, trained with billions of parameters to capture broad linguistic knowledge. Lightweight models, in contrast, are designed to minimize computational costs and facilitate deployment in resource-constrained environments. Finally, hybrid architectures combine transformers with other techniques, such as statistical models, alternative deep learning methods, or symbolic/linguistic resources, to further enhance analytical performance.

4.2 Datasets for sentiment analysis using transformers

To address the third research question (RQ3), which concerns the identification of key datasets in transformer-based sentiment analysis, this section presents and examines the most widely used datasets reported in the reviewed literature. The discussion highlights the defining characteristics of

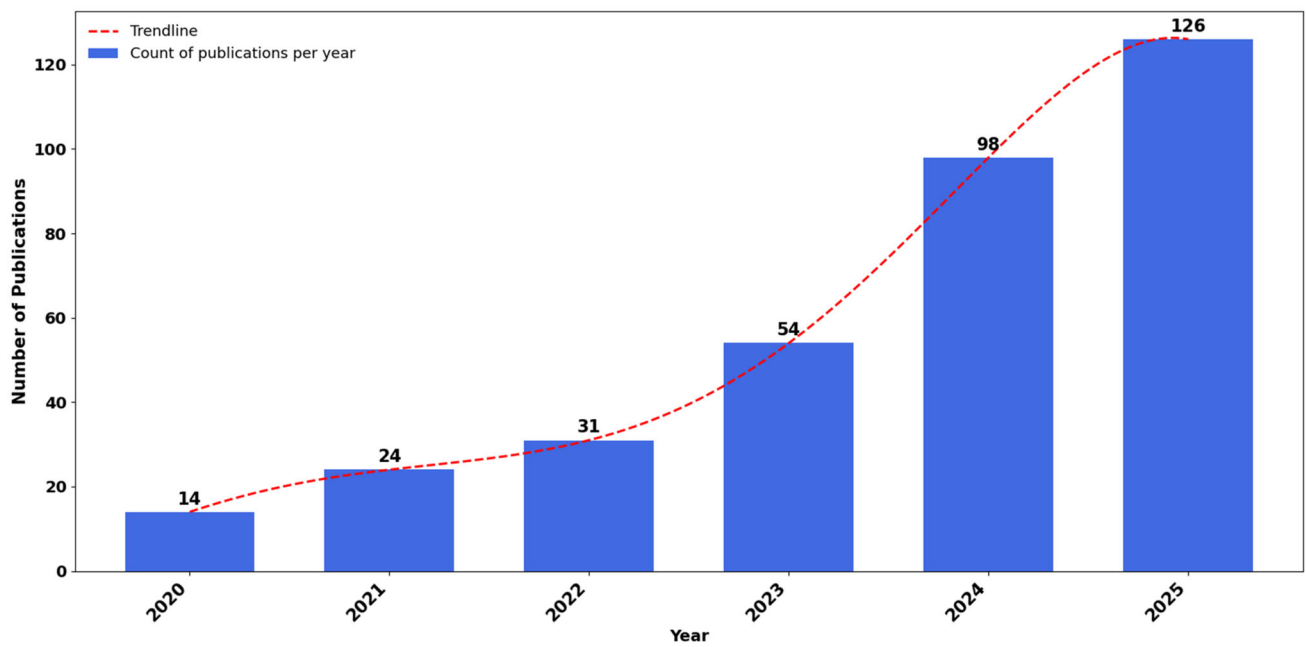


Fig. 4 The number of published papers on Sentiment analysis using transformers by year

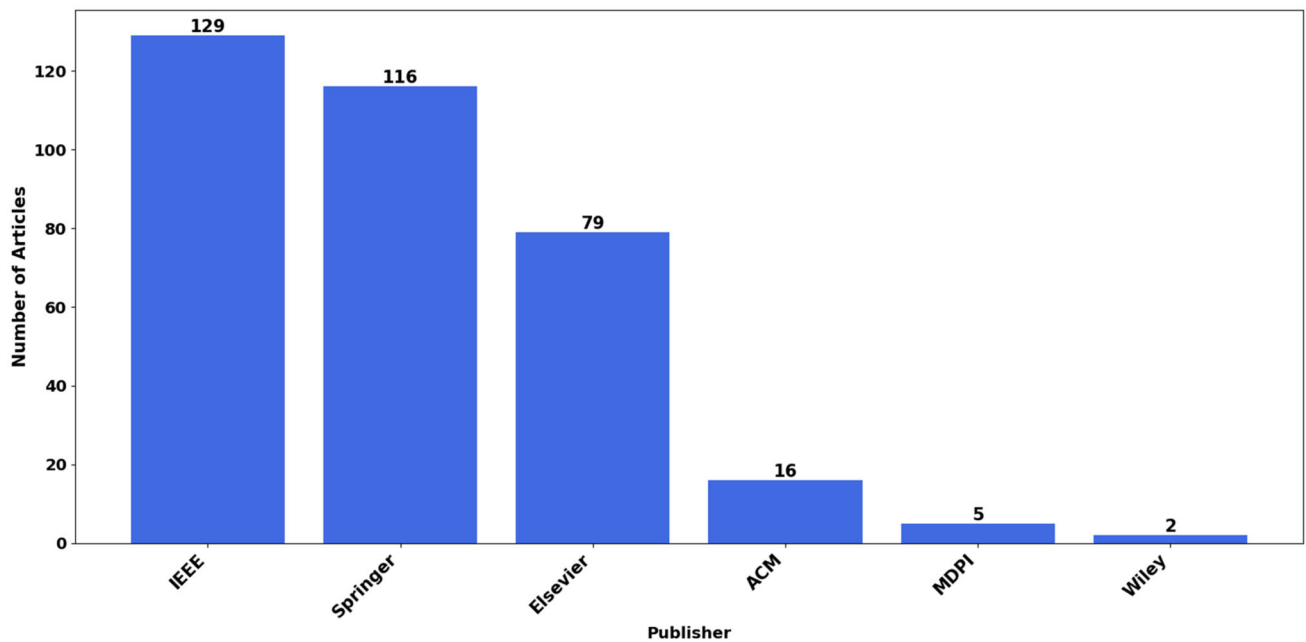


Fig. 5 The number of published papers on Sentiment analysis using transformers by publisher

each dataset, the frequency of their adoption across studies, and their relative significance within the broader landscape of sentiment analysis research.

4.2.1 Overview of widely used datasets

In the review of sentiment analysis studies, several key datasets have been identified. These datasets are summarized in Table 5 and discussed in more detail in Table 6.

Table 5 Widely-used datasets in sentiment analysis using transformers

Dataset name	Short description	Statistics	Used by
SemEval	SemEval datasets are among the most widely used benchmarks for sentiment analysis. Various versions from 2014, 2015, and 2016 have been utilized for sentiment classification in customer reviews, financial news, and aspect-based sentiment analysis	• SemEval-2014 Task 4 (Laptop & Restaurant): Laptop 14—2328 training, 638 test; Restaurant 14—3608 training, 1120 test • SemEval-2015 Task 12 (Restaurant): Restaurant 15—1315 training, 685 test • SemEval-2016 Task 5 (Restaurant): Restaurant 16—2000 training, 676 test. Also includes a 50,000 movie review dataset split evenly between training and testing	[18, 23, 40, 51, 53, 58, 70, 91, 100, 101, 109, 112, 114, 116–138]
IMDB	• The IMDB dataset consists of English-language movie reviews labeled as positive or negative. It is commonly used for binary sentiment analysis and, due to its diversity and length of texts, serves as a suitable benchmark for NLP models	Contains 50,000 movie reviews, evenly divided into 25,000 training and 25,000 testing samples	[52, 64, 65, 80, 111, 115, 139–144]
CMU-MOSEI	• CMU-MOSEI is one of the largest multimodal datasets for sentiment analysis, comprising YouTube videos with aligned text, visual, and audio modalities. Sentiments are labeled on a continuous scale from negative to positive and are useful for multimodal and NLP tasks	Contains 23,453 video clips from 5,000 videos, with sentiment labels ranging from -3 to +3	[41, 42, 45, 61, 63, 95, 108, 145–159]
CMU-MOSI	• CMU-MOSI consists of YouTube vlog videos where individuals express opinions about movies. It is designed for multimodal sentiment analysis with textual, audio, and visual data, labeled on a scale from negative to positive, and widely used in NLP and computer vision research	Includes 2,199 video clips of YouTube movie reviews. Sentiments are labeled between -3 and +3, including textual, audio, and visual features for multimodal sentiment analysis	[41, 42, 45, 61, 63, 108, 145–150, 152–159]

Table 5 (continued)

Dataset name	Short description	Statistics	Used by
Sentiment140	Sentiment140 contains tweets collected from Twitter and automatically labeled for sentiment analysis, comprising both positive and negative tweets. It is widely used in NLP and machine learning research for sentiment classification	Contains 1.6 million labeled tweets from Twitter	[111, 139, 143, 160–163]
Twitter US Airline	This dataset includes tweets from users about US airlines. Tweets are labeled as positive, negative, or neutral. It is commonly used to analyze customer satisfaction and sentiment in the airline service domain	Contains 14,640 tweets related to US airlines	[160], [57, 106, 143]
Yelp	Yelp dataset consists of user reviews regarding restaurants, cafes, and various businesses. It includes textual reviews accompanied by sentiment ratings, used for sentiment analysis, opinion classification, and NLP modeling	Comprises 561,000 user reviews about restaurants, cafes, and businesses	[18, 96, 116, 139, 144]
FiQA	FiQA contains financial and economic text data collected from sources such as news articles, social media, and financial forums. It is used for sentiment analysis, market trend prediction, and NLP in the financial domain, including positive, negative, and neutral sentiment labels	Includes 8000 sentences from financial forums and 10,000 sentences from economic news	[19, 164–166]

Table 6 Key features of the datasets

Task name	Labels	Features	Application domains	Data type	Limitations	Biases
SemEval-2014 Task 4	Aspect terms, categories, polarity (positive, negative, neutral, conflict)	Sentence-level customer reviews annotated with aspect terms and/or categories	Aspect extraction and aspect-level sentiment classification	Customer review datasets with human annotations identifying target aspects and their sentiment polarity	Relatively small dataset size Limited to restaurant and laptop domains Annotation inconsistencies between annotators Simple aspect categories may not capture complex opinions	Domain-specific bias (only restaurants /laptops) Annotator bias in subjective labeling English-only bias Potential class imbalance in aspect categories
SemEval-2015 Task 12	Entity-feature pairs, aspect terms, polarity (positive, negative, neutral, conflict)	Sentences with opinion targets, categories, and polarity annotations; includes out-of-domain test data	Cross-domain aspect-based sentiment analysis (ABSA)	SE-ABSA15 dataset consists of full reviews, not isolated sentences, likely out-of-context	Limited domain coverage Annotation quality varies across languages Inconsistent evaluation metrics across subtasks Small training sets for some languages	Western cultural bias in sentiment expression Language-specific biases in translation Domain bias toward reviews Temporal bias (data from a specific time period)
SemEval-2016 Task 5	Aspect categories, opinion target expressions, polarity (positive, negative, neutral); multilingual	Sentences with aspect terms/categories and polarity; multilingual domains (restaurants, hotels)	Multilingual ABSA and cross-domain evaluation	Review sentences in multiple languages and domains	Limited to specific domains (restaurants, laptops, hotels) Annotation guidelines may be inconsistent Small dataset sizes for some languages Focus on explicit aspects only	Cultural bias in sentiment norms Demographic bias of annotators Platform-specific language patterns Genre bias (formal review language)
IMDB	Binary sentiment: positive, negative (balanced)	Full-length movie reviews (up to several hundred words)	Document-level sentiment classification; long text modeling; benchmark for binary sentiment	English movie reviews, 25 k labeled training and 25 k testing samples; includes unlabeled data	Binary classification only (positive/negative) Limited to the movie domain Long text reviews may contain mixed sentiments Potential label noise from automated extraction	Temporal bias (older movies overrepresented) Genre bias in movie preferences Demographic bias of reviewers Cultural bias toward Western cinema
CMU-MOSEI	Sequential/continuous sentiment intensity: usually from -3 to +3 (or real-valued within [-3, +3]); often mapped to negative, neutral, positive classes	Short video segments with aligned transcripts, audio and visual features; timing and speaker segments	Multimodal sentiment analysis; regression and classification of sentiment intensity	Multimodal (text, visual, audio)	Multimodal complexity increases annotation difficulty Limited number of speakers Video quality variations affect analysis Expensive to scale and replicate	Speaker demographic bias Cultural expression bias Topic selection bias Recording environment bias

Table 6 (continued)

Task name	Labels	Features	Application domains	Data type	Limitations	Biases
CMU-MOSI	Sentiment [3] (continuous, 7-class, or binary)	Dataset with fine-grained annotations for subjectivity, sentiment intensity, visual features per frame and opinion, and audio features per millisecond	Multimodal sentiment and emotion recognition; robustness and large-scale benchmarking	Multimodal dataset of product reviews across various domains	Small dataset size Limited speaker diversity YouTube video context may not generalize Annotation subjectivity in multimodal fusion	Platform bias (YouTube-specific) Age and demographic bias of speakers Topic bias toward product reviews Temporal bias in video content
Sentiment140	Positive, negative, neutral	Tweets with distant supervision (emojis); user, timestamp, tweet text	Large-scale Twitter sentiment analysis; pretraining/weak supervision; domain adaptation; public opinion mining; lexicon development	English tweets with noisy/distant labels	Emoticon-based labeling may be noisy Limited context in tweets (140 chars) Automatic labeling reduces accuracy High class imbalance	Platform bias (Twitter-specific) Demographic bias of Twitter users Temporal bias (2009 data) Geographic bias toward English-speaking countries
Twitter US Airline	Positive, negative, neutral (manually labeled)	Tweets directed at US airlines; includes airline names, timestamps, and sometimes reason labels	Customer service sentiment analysis; social media monitoring; aspect/error analysis in tweets	English tweets, manually labeled	Highly domain-specific (airline complaints) Small dataset size Unbalanced classes (mostly negative) Limited temporal coverage	Strong negative sentiment bias US-centric airline experience bias Customer complaint bias Demographic bias of airline passengers
Yelp	Polarity: positive vs negative (Yelp Polarity); full: 1–5 star ratings (ordinal, Yelp Full)	Business reviews with titles and bodies; sometimes includes business metadata (category, location)	Document-level sentiment analysis; ordinal regression; domain-specific review modeling	English Yelp reviews, large-scale text	Business review domain limitation Star rating may not align with text sentiment Review length varies significantly Potential fake reviews	Geographic bias toward US businesses Socioeconomic bias of reviewers Business type bias Temporal bias in review patterns
FIQA	Sentiments: usually –1, 0, +1 or continuous polarity scores; also aspect categories (e.g., company, market, economy); includes tasks beyond sentiment (question answering, entity extraction)	Headlines, financial microblogs, statements with aspects and polarity; sometimes with reasons	Financial sentiment and aspect detection; downstream tasks such as volatility prediction	Financial domain: news, microblogs (English)	Limited to financial domain Complex financial jargon Small dataset size Requires domain expertise for annotation	Financial literacy bias Professional vs. retail investor bias Temporal bias (specific market periods) Geographic bias toward specific markets

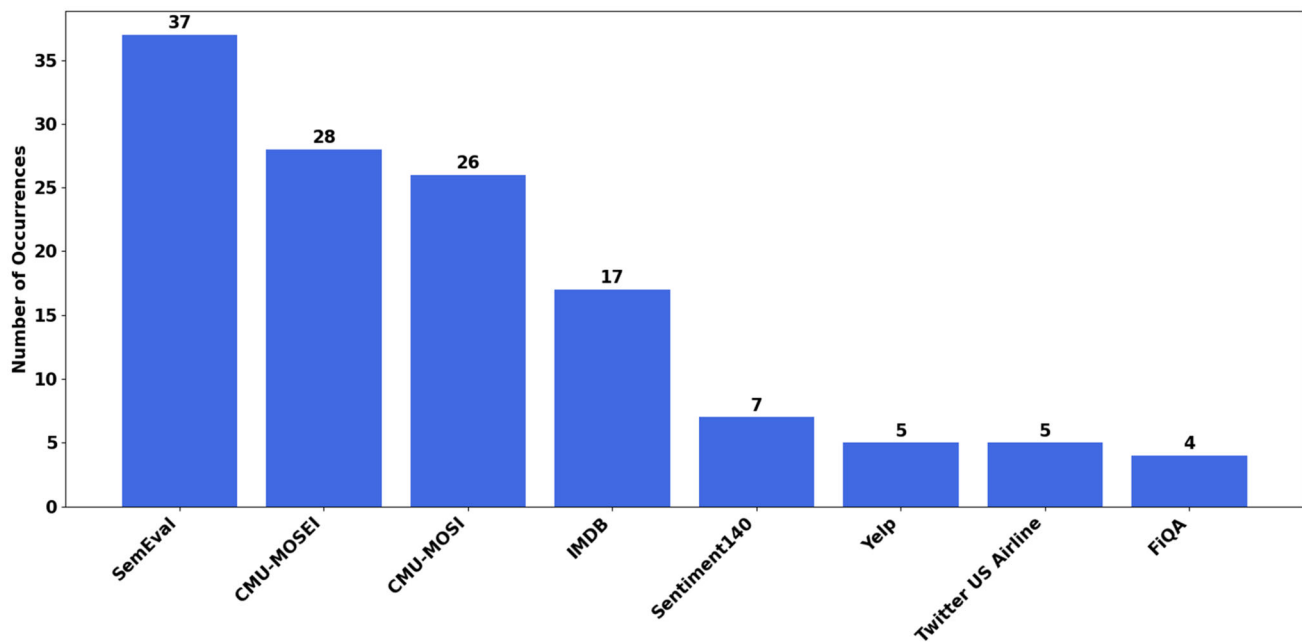


Fig. 6 Distribution of widely used datasets in different papers

4.2.2 Analysis of dataset distribution

As illustrated in the graph, the SemEval dataset is the most widely used, appearing in 36 studies. This highlights the role of SemEval competitions as a well-established benchmark for evaluating sentiment analysis approaches. Following this, the multimodal dataset CMU-MOSEI was used in 23 studies, CMU-MOSI in 21 studies, and the IMDB dataset in 17 studies. Figure 6 shows the number of times that each dataset was referenced by articles. This figure only shows the datasets that were referenced more than once.

4.2.3 Key features of datasets

Table 6 compares label schemas, features, application domains, data types, and explicitly reported limitations and biases for each dataset.

Based on the analysis summarized in Table 6, several recurring biases and limitations can be identified across the reviewed datasets:

Biases

- **Domain-specific bias:** Datasets limited to narrow domains (e.g., only restaurants, laptops, airlines, or financial discussions).
- **Language bias:** Predominantly or exclusively English-language data, with potential additional biases from translation processes.

- **Geographic and cultural bias:** Overrepresentation of English-speaking countries, especially the US, leading to Western/US-centric sentiment norms and expressions.
- **Demographic bias in reviewers/users:** Skewed representation of reviewers, Twitter users, airline passengers, or speakers by age, demographics, or socioeconomic status.
- **Annotator demographic and subjective bias:** Biases introduced by annotators' demographics or subjective judgments in labeling sentiment or aspects.
- **Sentiment class imbalance:** Datasets heavily skewed toward negative sentiment (e.g., complaints dominate, leading to unbalanced positive/neutral/negative classes).
- **Aspect category imbalance:** Uneven distribution of aspect categories in aspect-based sentiment analysis datasets.
- **Platform-specific bias:** Data influenced by the characteristics of specific platforms (e.g., Twitter complaints, YouTube comments).
- **Customer complaint bias:** Overrepresentation of negative experiences due to users being more likely to post complaints than positive feedback.
- **Specialized user bias:** Differences in perspective or literacy (e.g., professional vs. retail investors in financial sentiment data).

Limitations

- **Limited dataset size:** Many datasets are relatively small overall or have small training sets for certain languages/subtasks.

- **Restricted domain coverage:** Datasets are often confined to specific domains (e.g., restaurants, laptops, hotels, movies, airlines, finance), reducing generalizability.
- **Temporal bias and limited coverage:** Data often comes from specific time periods, with overrepresentation of older content or particular market/review patterns.
- **Annotation inconsistencies and quality issues:** Variations between annotators, across languages, or due to subjective guidelines lead to unreliable labels.
- **Label noise and misalignment:** Issues from automatic/emoticon-based labeling, fake reviews, or a mismatch between star ratings and textual sentiment.
- **Oversimplified sentiment/aspect modeling:** Focus on explicit aspects, binary polarity, or simple categories fails to capture complex, mixed, or nuanced opinions.
- **Lack of diversity in content/speakers:** Limited speaker numbers, diversity, recording environments, or cultural/genre representation (e.g., Western cinema bias).
- **Platform- or genre-specific biases:** Language patterns tied to platforms (e.g., short tweets), formal review styles, or topic bias toward product reviews.
- **Multimodal and technical challenges:** Increased annotation difficulty, video quality variations, or complexity in fusing modalities.
- **High cost and scalability issues:** Annotation requires domain expertise, is expensive to scale, and datasets (e.g., YouTube videos) may not generalize well.

As illustrated in Fig. 6 and Table 5, our systematic review identifies the following as the most dominant datasets:

- **SemEval (2014, 2015, 2016):** The most frequently used benchmark (appearing in 36 studies).
- **CMU-MOSEI & CMU-MOSI:** The standard benchmarks for multimodal analysis (used in 23 and 21 studies, respectively).
- **IMDB:** Widely used for long-text binary classification (17 studies).
- **Sentiment140:** A key dataset for large-scale, noisy social media text (10 studies).

In addition, we explicitly detail the distinctions among these datasets to guide researchers in selecting the most suitable benchmark for their specific task:

- For Aspect-Based Sentiment Analysis (ABSA): SemEval datasets are the primary choice because they provide sentence-level annotations with specific aspect terms and categories, making them suitable for fine-grained analysis tasks.
- For Multimodal Analysis: CMU-MOSEI and CMU-MOSI are distinct in that they include aligned audio, visual, and

textual data labeled on a continuous intensity scale (− 3 to + 3). They are suitable for testing transformer models' ability to fuse heterogeneous data streams.

- For Long-Document Modeling: IMDB consists of full-length movie reviews (up to several hundred words). It is uniquely suitable for evaluating a transformer's capacity to handle longer sequences and document-level context compared to short tweets.
- For Noisy/Informal Text: Sentiment140 and Twitter US Airline datasets are characterized by short, noisy texts containing emojis and slang. They are used to test model robustness in social media domains.
- For Specialized Domains: We highlighted FiQA as a key benchmark for the financial domain, suitable for tasks requiring the interpretation of economic news and microblogs.

4.3 Methods of sentiment analysis using transformers

In response to the second research question (RQ2), which focuses on identifying transformer-based models employed in sentiment analysis, the systematic review indicates that these models can be categorized into four principal groups. This taxonomy was designed to minimize conceptual overlap while ensuring comprehensive coverage of the models examined. The four categories are as follows:

- **Contextual-adaptive models:** This group includes models that are either pre-trained on diverse domains and subsequently fine-tuned for sentiment analysis within a specific domain, or purpose-built with tailored architectures and configurations for domain-specific data (e.g., financial, social, or medical datasets). The primary aim of this approach is to align the model with the distinct characteristics and requirements of each domain, thereby enhancing both accuracy and efficiency. Crucially, this adaptability extends beyond topic domains to encompass linguistic contexts. A critical subset of adaptive models focuses on linguistic diversity, treating language as a primary context. Multilingual transformers like mBERT and XLM-RoBERTa have become essential for handling morphological complexity in low-resource languages and code-switching (e.g., mixing Hindi and English) in social media text.
- **Large Language Models (LLMs):** These models, built upon transformer architectures with billions of parameters, possess the capacity to understand and process language at scale. Through techniques such as in-context learning, few-shot learning, and prompt-based strategies, LLMs can perform sentiment analysis at highly complex and diverse

levels. Their flexibility and generalizability enable their application across multiple domains and languages.

- **Lightweight models:** Designed to reduce computational complexity and optimize performance in resource-constrained settings, these models employ strategies such as model compression, pruning, and knowledge distillation. The goal is to minimize computational demands while preserving acceptable levels of accuracy. Such models are particularly well-suited for scenarios requiring real-time processing or deployment on hardware with limited resources.
- **Hybrid models:** In this category, transformer models are combined with complementary approaches and resources to enhance performance. This integration may involve other deep learning architectures (e.g., recurrent or convolutional networks), statistical methods, or linguistic and symbolic resources such as semantic lexicons, knowledge bases, or ontologies. The objective of these hybrid approaches is to harness the strengths of distributed representation learning alongside structured or conceptual knowledge to achieve more robust sentiment analysis.

As shown in Fig. 7, the approaches surveyed in this paper can be classified into a hierarchical taxonomy that organizes them according to their underlying principles and technical mechanisms.

Figure 8 illustrates the distribution of published studies across these categories, while Fig. 9 depicts the yearly trends and growth rates associated with each approach.

In the following section, each category is studied in depth to provide a clearer picture of the current state of transformer-based models for sentiment analysis.

4.3.1 Contextual-adaptive models

Fiok et al. [17] compiled a sentiment dataset, known as sentiment@USNavy, from Twitter posts directed to a single U.S. Navy user account. They applied explainable AI techniques to transformer models for Twitter sentiment analysis and investigated whether topics extracted through Latent Dirichlet Allocation (LDA) could enhance the accuracy of these transformer models. However, the study faced limitations, including a lack of parameter optimization for feature extractors that introduced potential bias, reliance on a single machine learning classifier (gradient boosting) across all models, and the use of a relatively simple text preprocessing pipeline that may have negatively affected the performance of newer transformer architectures.

Souza et al. [16] tested a pretrained approach using thirteen different aggregation strategies for BERT token vectors, which were then passed to a logistic regression classifier, alongside a fine-tuning approach that added a linear layer on the [CLS] output and employed the Adam optimizer. They

experimented with both BERTimbau, a Portuguese-specific model, and multilingual BERT (m-BERT). The limitations included the pretrained method's dependence on a simple linear classifier, which restricted its ability to capture complex nonlinear relationships and may not have fully leveraged BERT's sophisticated feature representations, as well as the study's restriction to the Portuguese language context.

Karabila et al. [90] fine-tuned a BERT_BASE model on the Amazon Musical Instruments dataset for binary sentiment classification, then combined the resulting sentiment scores with collaborative filtering methods using support vector regression (SVR) to produce final recommendations as a weighted combination of predicted ratings and sentiment scores. Limitations of this work included its scope being restricted to a single domain, with untested generalizability to others, the dataset suffering from unaddressed class imbalance issues, and the sentiment analysis component requiring more advanced algorithms and complementary data sources.

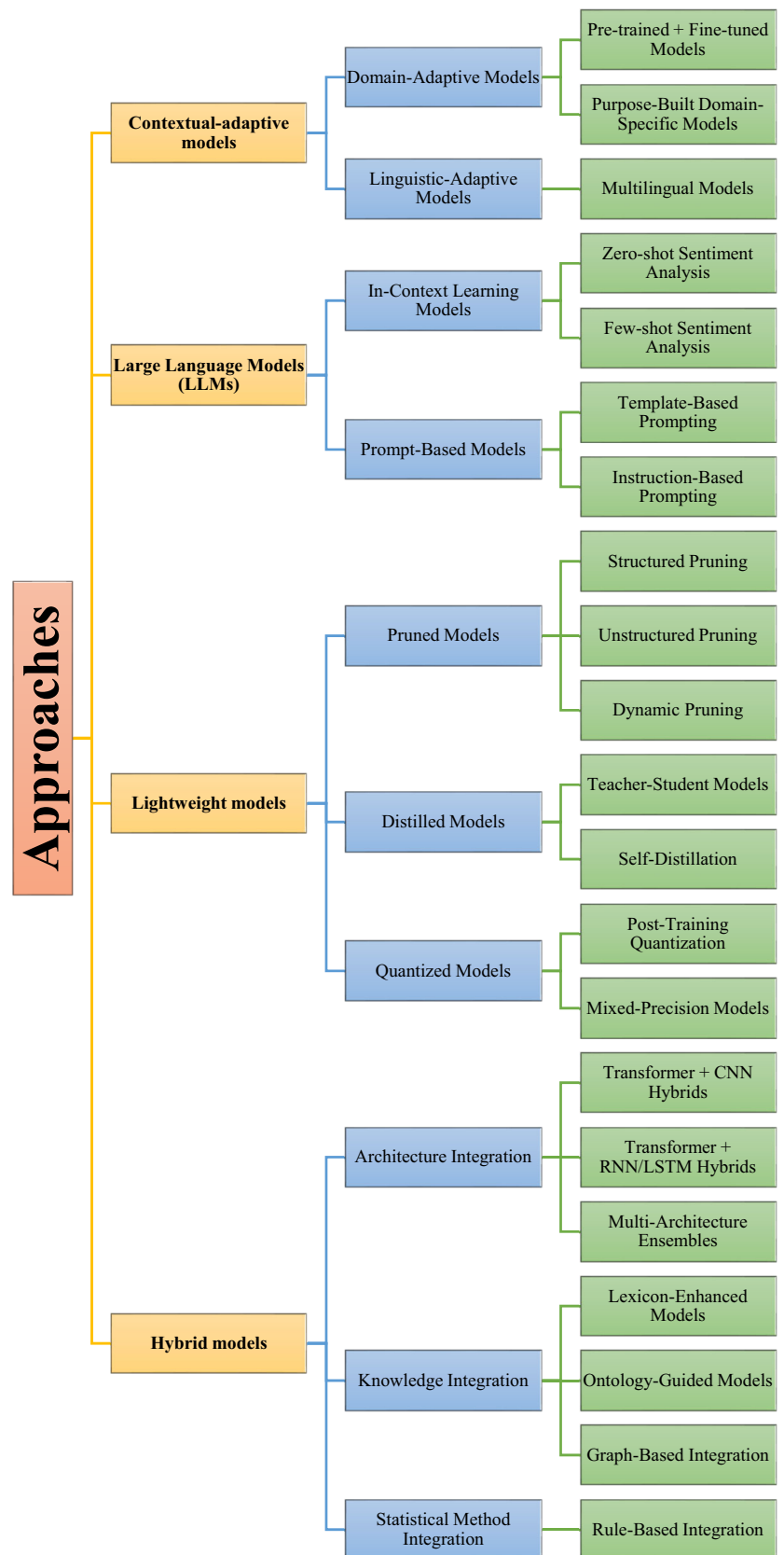
Chauhan et al. [58] employed a BERT sentence pair classification (SPC) model for aspect-based sentiment analysis, feeding input sentences into BERT layers, combining embeddings through fully connected layers, and applying aggregation layers for the final classification of sentence pairs. The approach showed limitations in the SPC model's reliability when evaluating opinions in certain datasets, struggles with the inherently informal and noisy nature of social media text, and persistent challenges in the aspect-based sentiment analysis setting overall.

Murtadha et al. [91] proposed a unified framework that used supervised LDA to extract seed terms for implicit aspects, generated auxiliary sentences by modeling semantic relations with corresponding aspect seeds, and fine-tuned BERT to capture aspect-specific representations based on this auxiliary input. Limitations arose from the dependency parser producing syntactic errors that affected accuracy, incorrect semantic candidates when tokens were erroneously associated with unrelated aspect seeds, and incomplete coverage in cases where no suitable semantic candidate was identified.

Lee et al. [81] employed RoBERTa to overcome the limitations of traditional rule-based sentiment analysis and adopted BERTopic for topic modeling using pretrained BERT embeddings, encoding documents into high-dimensional embeddings to capture semantic and contextual information. The study was limited by a sample restricted to teachers and non-teachers that may not represent broader educator perspectives, BERTopic's ongoing challenges with very short texts despite improvements over LDA, and the possibility that automatic topic determination might not always align with human interpretation.

Vazirani et al. [92] used the NLTK sentence tokenizer to segment central bank speeches into individual sentences

Fig. 7 Taxonomy of sentiment analysis methods using transformers



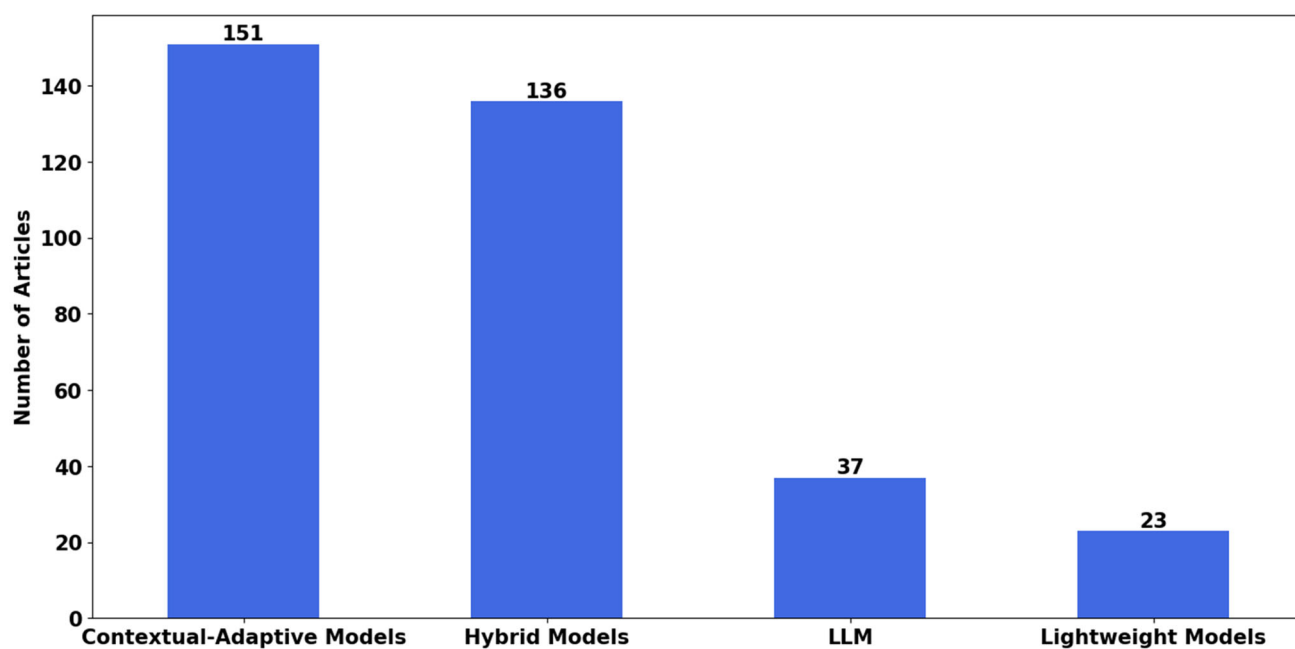


Fig. 8 Number of published articles in each category

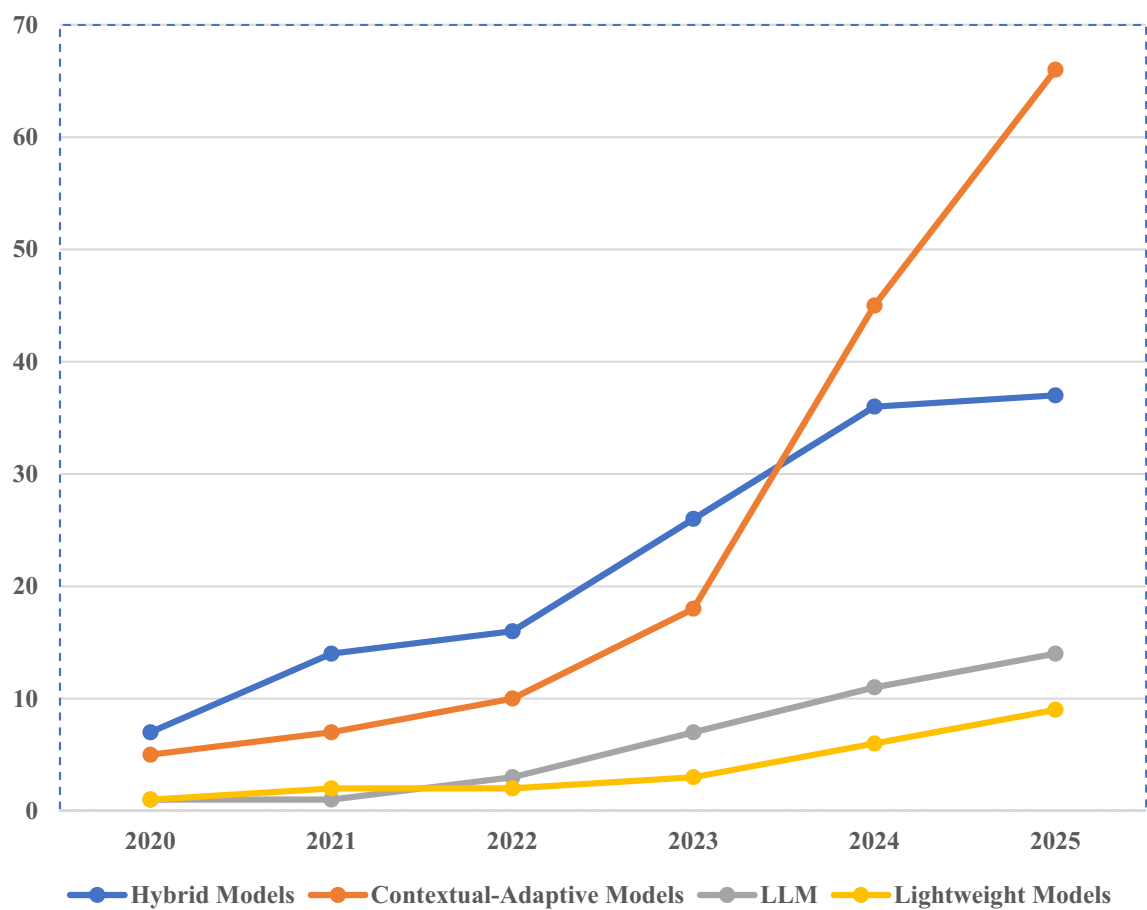


Fig. 9 Trend line of published articles in each category by year

and applied two fine-tuned RoBERTa models—a specialized financial model and the general SiBERT model—to analyze both financial and general sentiment across multiple countries. Potential limitations included inaccuracies in the data-cleaning process that could bias results, constraints on country and speech data selection due to availability and model compatibility, and the omission of other influential factors such as technological development.

Chandra et al. [39] implemented a sequential framework involving data selection, cleaning, n-gram visualization, and model fine-tuning, applying BERT and RoBERTa to Guardian articles for multi-label emotion classification while using visualization and comparative analysis across different time periods. Limitations stemmed from the models being fine-tuned on the SenWave dataset, which contained social media texts with abbreviations, slang, and emojis that created challenges for robust training, as well as a domain mismatch between the social media training data and the news article application data.

Rastogi et al. [93] proposed an automated annotation strategy using weak supervision through the Snorkel system, translating multilingual data into English, defining six labeling functions, and applying BERT-base-uncased and RoBERTa to both translated and original datasets. The approach was limited by the Snorkel framework's lack of support for multilingual data, necessitating a translation preprocessing step that could introduce errors and lose language-specific nuances, alongside the risk of the weak supervision approach propagating labeling errors throughout the system.

Hoque et al. [47] evaluated five pretrained transformer models with hyperparameter tuning on a Bengali dataset and proposed a Transformer-Ensemble hybrid model integrating mBERT, BanglaBERT, and XLM-R-base, applying different tokenization techniques such as WordPiece for some models and SentencePiece for XLM-R. Limitations included underperformance on Bengali-English code-mixed data, a dataset with relatively few negative samples that affected negative class identification, and the overall limited resources available for Bengali sentiment analysis research.

Goud et al. [40] developed HybBERT, integrating BERT, RoBERTa, and DistilBERT for aspect-based sentiment analysis, preprocessing the SemEval 2014 data, evaluating models individually and in combinations, and performing joint aspect extraction and sentiment classification tasks. Some model combinations, particularly those involving RoBERTa, yielded weaker results, performance consistency across all setups could not be guaranteed, and the evaluation was limited to a single dataset, requiring further investigation for generalization.

Contreras Hernández et al. [94] collected Spanish COVID-19 tweets using the Twint library, applied lexicon-based labeling with manual correction, trained Spanish-specific BERT models and ten multilingual models on a balanced dataset, and performed preprocessing including character removal, case standardization, and dataset balancing. Limitations involved the lexicon-based labeling potentially misclassifying sarcastic or ironic expressions, restriction to a single topic (COVID-19 in Mexico) and time-frame that affected generalizability, and an exclusive focus on Spanish that limited applicability to other languages or dialects.

Lin et al. [38] applied COVID-Twitter-BERT (CT-BERT) with an Auxiliary Sentence strategy for emotion classification, exploring single-sentence and pair-sentence classification with 12 binary instances per sample using the SimpleTransformers library and five-fold cross-validation. CT-BERT did not outperform general models, likely due to non-English tokens in the dataset, the model was trained exclusively on English data, limiting multilingual performance, and no direct evaluation framework was defined to compare pair-sentence versus single-sentence effectiveness.

Mamta et al. [71] developed a shared-private multilingual framework with a shared mBERT encoder for task-general features and task-specific mBERT encoders for language-specific representations with joint fine-tuning, treating code-mixed sentiment as the primary task and English sentiment as auxiliary. The model struggled with implicit sentiment detection in noisy social media content, had difficulty handling sentences with mixed or conflicting emotions, and showed limited effectiveness on tweets with complex emotional expressions.

Chlapanis et al. [95] developed Adapted Multimodal BERT (AMB) with adapter layers for task-specific fine-tuning, fusing textual, audio, and visual features through hierarchical feed-forward networks and implementing multimodal fusion across 12 layers to enable cross-modal interactions. The approach did not address temporal "shifting effects" between modalities, misalignment between textual and audio-visual modalities remained unresolved, and the adapter-based method may not have captured all necessary cross-modal dependencies.

Riaz et al. [46] developed Twitter-Modified BERT (TM-BERT) with modified hyperparameters for COVID-19 vaccination tweets, pretrained on a combination of Wikipedia and Twitter data with a streamlined architecture, and fine-tuned models on a constructed CV-SAT labeled dataset. Evaluation was restricted to COVID-19 vaccination tweets, limiting domain generalizability, there was uncertainty about performance across other languages or social media platforms, and reliance on Twitter data introduced vulnerability to noise and bias during pretraining.

4.3.2 Large language models

Chumakov et al. [18] proposed a domain-independent generative approach for aspect-sentiment-triplet extraction using pretrained GPT models. They applied few-shot learning by prepending balanced examples from relevant domains before the input text and fine-tuned smaller GPT-2 models in a sample-by-sample manner for processing both English and Russian text. However, the smaller RuGPT-3 models were unable to perform tasks under the few-shot paradigm and required fine-tuning instead. Few-shot learning proved effective only for very large and resource-intensive models like GPT-3, and the method's overall effectiveness was limited by model size constraints that affected practical deployment.

Deng et al. [19] developed a semi-supervised framework that utilized a large language model (PaLM-540B) to generate weak sentiment labels for Reddit financial discussions, enhancing performance through chain-of-thought reasoning by producing eight generations per input to capture diverse reasoning paths, and then trained a student model via knowledge distillation using a regression-based loss function that aggregated multiple LLM predictions. Limitations included the key challenge of ensuring responsible integration of model outputs without misleading users or amplifying harmful behaviors, uncertainty about whether human annotators could effectively contribute through domain-knowledge-enriched prompt design, and remaining questions about the ability of automated prompt tuning to optimize human-engineered prompts effectively.

Benayas et al. [43] conducted comparative experiments between the encoder-only RoBERTa model and the decoder-only LLaMA2 for intent classification and sentiment analysis, fine-tuning RoBERTa with an additional classification layer while applying LoRA and QLoRA techniques to LLaMA2, and exploring zero-shot and few-shot learning approaches to improve generalization across varying intent formulations. The study was limited by its exclusive focus on LLaMA2 as the sole representative of decoder-only models, dependence on public benchmark datasets that may lack the diversity of proprietary real-world data, and limited attention to optimizing input prompt design, which could have impacted model performance potential.

Krugmann et al. [96] empirically examined the performance of GPT-3.5, GPT-4, and Llama 2 in binary and three-class sentiment classification under zero-shot settings, investigating the influence of data characteristics and different prompting strategies on classification accuracy across more than 20 datasets, and assessing model output interpretability through a user study involving marketing experts without any fine-tuning or parameter optimization. Performance declined significantly on noisy data sources and in multi-class classification tasks; outcomes were highly sensitive to prompt design, which complicated reproducibility, and

the study focused exclusively on off-the-shelf models without exploring potential improvements through fine-tuning.

Scerri et al. [44] applied ChatGPT-4 for sentiment analysis of user feedback in a Virtual Urban Living Lab platform, using a standardized direct prompting template followed by a numbered listing of user comments, and manually reviewed the results to confirm no hallucinations occurred while verifying the LLM's sentiment analysis accuracy. Limitations included potential difficulties in interpreting technical or culturally specific language within the urban planning context, the presence of output biases in ChatGPT that affected sentiment evaluation reliability, and limited validation beyond manual review that could miss subtle interpretation errors.

Radaideh et al. [97] investigated social bias using five LLMs (BERT, GPT-2, LLaMA-2, Falcon, and MistralAI) fine-tuned on social media datasets, conducting 1,500 pattern-based experiments with simple and complex input patterns across different topical domains, and applying explainable AI methods to analyze how bias-associated words contributed to sentiment prediction outcomes. The reliance on formal, non-sarcastic, and non-toxic input patterns may have underestimated real-world fairness concerns. The study used only 100 examples per bias category, which limited statistical power and generalizability, and there is a need for more fine-grained analyses in future research to capture subtle bias manifestations.

4.3.3 Lightweight models

Putrada et al. [20] designed a DistilBERT-based sentiment analysis model featuring a fixed 20-token input layer, a pre-trained DistilBERT layer, two fully connected layers with 768 and 512 neurons, respectively, and a Softmax output. They applied a preprocessing pipeline that included tokenization, tensor generation, sampling, and DataLoader construction on the Rotten Tomatoes dataset using 1,000 samples, then trained the model with the AdamW optimizer and compared its performance against baselines such as SVM + TF-IDF, NB + TF-IDF, and CNN. Limitations included the small dataset size of 1,000 samples, which restricted model generalizability to broader applications, a fixed input length of 20 tokens that risked truncating longer reviews and discarding valuable semantic information, and the limited dataset size, potentially failing to capture sufficient linguistic diversity for robust performance evaluation.

Moura et al. [21] proposed a structured pruning strategy for BERTweet self-attention heads, using accumulated gradient values during fine-tuning to rank head importance, and applied iterative pruning with different search strategies, such as greedy search and hill climbing, to select the optimal architecture. They evaluated the approach across 22 Twitter datasets in two independent phases with multiple repetitions using different random seeds. The method was

limited in that it was only tested on BERTweet for tweet sentiment analysis, raising questions about generalizability to other models or NLP tasks, the head-importance ranking and optimal-architecture search required multiple repetitions with different seeds, making it computationally expensive and hardware-intensive, and these resource demands limited feasibility in environments with restricted computational capacity.

Ng et al. [64] employed DistilBERT with WordPiece tokenization, sentence padding, truncation, and class weighting on the IMDB and Customer Reviews datasets. The architecture included an embedding layer, six encoder layers with multi-head self-attention and feed-forward networks, and a classification layer on the [CLS] token, with training involving freezing lower encoder layers, learning rate scheduling, early stopping, and random search hyperparameter tuning. Despite solid performance, improvements over prior baselines were relatively modest, reliance on only two relatively small and simple datasets constrained generalizability to more complex domains, and the limited dataset diversity may not have adequately tested model robustness across varied linguistic patterns.

Salmony et al. [65] introduced a knowledge distillation framework using BERT as the teacher model and lightweight algorithms such as Naive Bayes, SVM, and Logistic Regression as students. They applied Bag-of-Words with an n -gram range of 1 to 3 for the students and BertTokenizer with 512 tokens for the teacher, training the students with soft labels derived from BERT's logits on 2,000 randomly selected samples from the IMDB dataset due to computational constraints. Limitations arose from a fundamental mismatch between BERT's contextual high-dimensional representations and the students' statistical Bag-of-Words features that prevented full knowledge transfer, the dataset reduction from 50,000 to 2,000 reviews which narrowed linguistic diversity and increased overfitting risk, and the limited sample size constraining the model's ability to learn comprehensive semantic patterns.

Ghorpade et al. [98] compared five deep learning architectures—Simple RNN, LSTM, GRU, Bi-Directional RNN, and BERT via DistilBERT—on toxic comment classification. They applied preprocessing by merging toxic categories into a binary class, followed by tokenization and padding on 223,549 English comments, and trained all models under identical hyperparameter settings on GPUs for a fair comparison. The study was limited by its reliance on an exclusively single-language dataset (English) and binary classification, restricting applicability to other languages and domains, evaluation based on a single data source without exploring the influence of varied parameters or dataset diversity, and a lack of multi-domain testing that restricted broader conclusions about model generalizability.

4.3.4 Hybrid models

Ouyang et al. [99] proposed a cross-domain Mean Teacher framework for aspect-based sentiment analysis, featuring two identical networks, each containing student and teacher components with the same architecture. They applied data augmentation by randomly replacing words or phrases with synonyms to introduce diversity between modules, training one on original data and the other on augmented data, then combined data from both domains to enhance the student network's task classifier using cross-entropy loss for labeled data and pseudo-labels from the teacher for unlabeled target domain data. Limitations included high memory usage due to joint training of two student–teacher modules requiring substantial computational resources, limited diversity in augmented samples as simple random replacement did not yield sufficiently varied examples, and framework sensitivity to pseudo-label quality that may disrupt the extraction of domain-invariant features.

Ahmad et al. [100] developed the EAFL-ACSA method, integrating federated learning with a Balance Augmentation Mechanism for aspect category sentiment analysis, combining privacy-preserving capabilities with diverse augmented data representations to provide multiple complementary features, and incorporating an efficient parameter-tuning component to improve model performance and generate refined augmented sentences. The approach faced limitations in pattern alignment, particularly with labeled word representations that limited framework effectiveness, a training strategy requiring substantial computational resources due to reliance on fine-tuning large pretrained models, and occasional production of incorrect polarity representations by random sentence generation, indicating the need for more sophisticated augmentation strategies.

Dong et al. [101] introduced a rule-based static optimization using handcrafted priority of dependency relations to reorder context, developed a score-based dynamic optimization calculating word position scores using a neural network to regulate contextual sequence, and proposed the PGSO network with prompt construction for task-specific prompts and a sequence regulator leveraging semantic, syntactic, and original-sequence information. Limitations involved algorithms relying on representation scores, sometimes overlooking information in the original sequence, dynamic optimization potentially resulting in incorrect predictions due to information loss, and framework complexity requiring careful balance between optimization benefits and computational overhead.

Yan et al. [102] combined a BERT model with a Sentiment Knowledge Lexicon containing positive, negative, intensifier, and negation words for aspect-level sentiment classification, generating static vectors from the BERT embedding layer and dynamic vectors from the BERT model for input

text and sentiment keywords, then applying a multi-head attention mechanism to produce four weighted representations for different sentiment word categories plus a core text representation. The restricted size of the Sentiment Knowledge Lexicon may not cover the full spectrum of sentiment words, especially in domain-specific texts. The model was highly sensitive to lexicon coverage and quality, limiting performance in diverse linguistic contexts, and dependency on a predefined lexicon constrained adaptability to new domains or evolving language patterns.

Sheetal et al. [103] integrated the Loughran-McDonald dictionary with BERT for financial sentiment analysis, using the dictionary for domain-specific word labeling while BERT captured nuanced meanings and semantic relationships in financial texts, and created a PyTorch dataset with BERT tokenization processed through the model using dropout regularization and cross-entropy loss optimization. The model began overfitting after 5 epochs, indicating the need for earlier stopping or additional regularization techniques. Evaluation was limited to a single domain (financial), raising questions about generalizability to other text types, and the integration approach may not have fully leveraged potential synergies between traditional and modern NLP methods.

Li et al. [22] leveraged a BERT architecture pre-trained on Chinese Wikipedia to generate sentence representations using the [CLS] token for classification, designed and evaluated three alternative architectures (BERT + FC, BERT + LSTM, BERT + CNN), with BERT + FC selected for superior performance, and fine-tuned the model on 9,204 balanced Chinese stock reviews collected from the Guba platform. The relatively small volume of labeled data limited the model's ability to learn comprehensive patterns, restriction to single dataset testing constrained generalizability across different domains, and the limited validation scope required testing on larger or cross-domain datasets for robust performance assessment.

Zhu et al. [23] employed an automatic prompt-pattern generation method to identify effective sentiment keywords and templates for each dataset, applied a multi-prompt learning strategy to expand training data in a targeted manner addressing class imbalance issues, and implemented Global Semantic Refinement using multi-layer pointwise convolutions alongside Multi-grained Local Semantic Extraction with parallel CNNs. Model performance was highly dependent on the quality and diversity of prompt templates and training data, the automatic prompt search and ranking process was computationally intensive, requiring substantial processing resources, and framework scalability was limited by resource demands, making it less practical for widespread deployment.

Zhang et al. [104] applied a BERT Word Encoding layer for semantic word representations, followed by a Transformer Semantic Extraction for deep features, employed

an Interactive Attention layer to strengthen mutual alignment between aspects and review text, enhancing contextual information, and used a Sentiment Classification layer for polarity prediction with the model trained on labeled datasets. The model did not incorporate location information, which has been shown to improve classification accuracy in prior studies; omitting positional features represented a missed opportunity for performance enhancement, and the framework may not have fully captured spatial relationships between aspects and sentiment expressions in text.

Liu et al. [105] utilized a BERT dynamic pre-training language model to encode text, enabling rich semantic features and contextual understanding, employed stacked involutions with varying dimensions to extract features and perceive local information across different scales, and incorporated nonlinear activation functions (ReLU6 and H-Swish) to enhance model expression capability and performance. High computational complexity of pre-training and stacked network models limited practical deployment; the increased network depth approach may still suffer from optimization challenges despite involution improvements, and the resource-intensive architecture may not be suitable for applications with computational constraints.

Vaish et al. [106] developed a Dynamic Attention-Based Feature Selection method combined with a Genetic Algorithm to optimize attention weights and hyperparameters, applying the GA to optimize both feature selection attention weights and model hyperparameters like layers and learning rate, while integrating convolutional and pooling layers into BERT to extract more dataset-relevant features, creating an enhanced BERT variant. The approach suffered from relatively high runtime, which may hinder scalability in practical applications; the complex optimization process required substantial computational resources for GA-based parameter tuning, and the feature selection method may not generalize well across different datasets without extensive retraining.

Dubey et al. [107] combined dynamic knowledge graph construction with real-time contextual adaptation and hierarchical sentiment representation, applying Named Entity Recognition for sentiment-related objects and dependency parsing for aspect extraction with multi-relational graph edges, while implementing a Dynamic Adaptive Knowledge Graph updating with new data and temporal emotional modeling with time-enriched edges. Limited adaptability in specialized domains restricted framework application across diverse contexts, significant computational demands made the system resource-intensive for real-world deployment, and difficulties in tracking rapid sentiment changes limited effectiveness in dynamic environments.

Li et al. [108] proposed the CSMF-SPC model with a bidirectional long-term memory architecture capturing contextual semantics from non-textual modalities, applying BERT for textual encoding and a low-rank fusion method

to comprehensively model cross-modal interactions, while incorporating a weight self-adjusting polarity penalty loss function enabling the system to backpropagate additional sentiment features. The model was constrained by reliance on the presence of all three modalities failing when one or more were missing, the framework could not perform prediction without complete multimodal input, limiting practical applicability, and dependency on full modality availability restricted usage in scenarios with incomplete or missing data streams.

Singh et al. [74] integrated sentiment and situational features using the TA-lib library to extract financial indicators influencing stock prices, performed sentiment analysis using FinBERT for bidirectional text encoding, classifying sentiment as positive, negative, or neutral, and combined predicted polarities with historical stock data, testing bidirectional RNNs and Bi-LSTMs for news impact assessment. The narrow scope focusing solely on English-language news raised concerns about generalizability across heterogeneous contexts, limited linguistic diversity constrained model applicability to global financial markets with multiple languages, and single-language evaluation was insufficient for validating framework robustness across diverse cultural and linguistic contexts.

Hou et al. [42] developed a tensor-based framework, TF-BERT, converting textual data into vector representations through word embeddings and BERT transformer layers, integrating Tensor-based Cross-modal Fusion modules starting from specific transformer layers to enhance modalities through interactions, and applying a tensor-driven strategy extracting sentiment signals from source modalities within the TCF module for target modality enrichment. The model incurred substantial computational costs, making it less practical for low-power or real-time scenarios, the framework was unsuitable for deployment on mobile or edge devices due to resource requirements, and high computational demands limited scalability and practical adoption in resource-constrained environments.

He et al. [109] employed a frozen pre-trained BERT to extract word-level feature vectors, reducing overfitting and accelerating training, applied CABiLSTM preserving implicit sequence features bidirectionally across all layers with convolution operations for integration, and used multi-head self-attention for salient feature extraction followed by average pooling and a sentiment classification layer. The model was suboptimal in handling interactions between emotions across different aspects and levels during computation. Classification performance was highly sensitive to class distribution of the dataset, affecting robustness, and the framework may struggle with complex emotional interactions requiring more sophisticated attention mechanisms.

Chauhan et al. [110] integrated three transfer learning models (XLNet, BERT, RoBERTa) into a unified architecture with outputs leveraged by a LightGBM model, applying gradient boosting for improved performance, exploiting synergistic fusion of base models, and conducting extensive hyperparameter optimization while fine-tuning pre-trained BERT on a benchmark dataset, updating all layers. The approach faced scalability challenges, having not been evaluated on larger datasets or diverse platforms, performance remained untested in complex real-world scenarios like long-form text or multilingual data analysis, and exclusive reliance on English-language comments limited applicability across global linguistic contexts.

Subba et al. [111] generated three distinct embeddings (custom Word2Vec, pretrained GloVe, pretrained BERT) serving as input to base-level classifiers, applied LSTM, GRU, and Bi-GRU as base classifiers with frozen pretrained embedding layers preventing weight updates, and combined base model predictions to train a meta-level LSTM classifier using carefully designed data partitioning strategies. Reliance on hybrid RNNs with transformers limited the ability to capture intricate contextual dependencies of full transformer architectures; the framework may not effectively model complex relationships that advanced transformer-based systems can capture, and the architecture design prevented leveraging the full potential of modern transformer capabilities for sentiment understanding.

Phan et al. [112] encoded words into semantic vectors using pretrained BERT, then processed by BiLSTM for contextualized representations, applied a Graph Convolutional Network to model syntactic dependencies in graph form, extracted salient features, and used CNN layers on GCN output for feature refinement, followed by a fully connected layer with Softmax for classification. The model failed to simultaneously capture all contextual cues, semantic relations, and sentiment knowledge within the text graph. Performance was highly sensitive to input data quality and careful tuning of CNN and GCN layer configurations, and sensitivity to hyperparameter configuration increased overfitting risk on certain datasets, limiting generalizability.

He et al. [113] tokenized and transformed text into contextual embeddings using BERT, then passed through CNN layers with multiple kernel sizes, applied Max Pooling operations for local feature capture followed by a BiLSTM layer for bidirectional and long-range dependencies, and used a Dropout layer (0.5 probability) for overfitting reduction with final prediction through a fully connected Softmax layer. The model introduced significant architectural complexity and computational overhead compared to simpler alternatives. Additional CNN and BiLSTM layers may lead to redundancy since BERT already captures many dependencies, and redundant processing reduces efficiency in real-world applications despite achieving high performance.

Sivakumar et al. [52] integrated the multi-head attention mechanism of BERT with a Bidirectional GRU classifier through four main processing stages, applying WordPiece preprocessing, parsing-composition for syntactic trees, and an attention stage addressing polysemy and homonymy, then feeding BERT-derived features into Bi-GRU for joint modeling of temporal and semantic dependencies for sentiment prediction. The framework required considerable computational resources, especially when fine-tuning BERT, limiting accessibility. Performance may deteriorate on highly noisy datasets or across different language domains without domain-specific adaptation, and the model remained highly sensitive to preprocessing quality, affecting overall system robustness and reliability.

Guan et al. [114] constructed an aspect-dependent syntactic graph using aspect-specific masking encoded via a graph attention network to mitigate noise, built a semantic graph from self-attention weights from BERT higher layers processed using a Graph Convolutional Network, and applied supervised and unsupervised contrastive learning for representation discriminability and syntactic-semantic weighting optimization. The model struggled with implicit or nuanced sentiments lacking explicit cues, limiting effectiveness on subtle expressions, performed less reliably on data with informal or incomplete syntactic structures, such as social media tweets, and success heavily relied on BERT's capacity to capture subtle semantic meanings, constraining performance scope.

Ai et al. [45] independently extracted textual features using GloVe embeddings with Bi-GRU, visual features through 3D-CNN, and acoustic features via the COVAREP toolkit, projected each modality into a unified dimensional space via dense layers then concatenated features for Universal Transformer input, and applied the Universal Transformer for intermodal dependency modeling followed by a Softmax layer for final sentiment classification. Audio features were extracted at a shallow level, limiting expressive capacity and resulting in loss of rich speech characteristics, the framework may not capture the full complexity of acoustic information due to the basic feature extraction approach, and limited acoustic processing prevented leveraging advanced speech analysis techniques for improved multimodal understanding.

Jia et al. [115] processed input text through BERT for contextualized word embeddings, then parallel processing by TextCNN, BiLSTM, and BiGRU modules, applying TextCNN for local feature extraction while BiLSTM and BiGRU captured global dependencies across text sequences, and combined and reweighted module outputs using a self-attention mechanism followed by a multilayer perceptron with Softmax classification. The hybrid design significantly increased computational complexity and training time compared to baseline models. Fine-tuning parameters across multiple submodules was challenging and demanded substantial

computational resources, and the complex architecture may not provide proportional performance benefits relative to increased computational costs.

Vo et al. [116] built a neural-fuzzy architecture on a deep transformer encoder-decoder framework incorporating a topic-aware attention mechanism within the transformer backbone, modeling globally salient latent topics and rich contextual relations simultaneously to improve text representation quality, and feeding topic-based semantic embeddings into a hybrid fuzzy deep neural network, reducing noise and ambiguity before classification. The current version was not designed or evaluated for real-time or dynamic sentiment analysis tasks, the framework had limitations in handling streaming or interactive sentiment analysis scenarios, and the static design approach may not accommodate evolving sentiment patterns in dynamic conversational contexts.

Wang et al. [41] developed a Cross-modal Enhancement Network with a CE module augmenting word embeddings with long-range emotional signals from non-verbal data, introduced a feature transformation strategy for visual and audio features aligning their distributions for integration with textual features, and applied pre-trained embeddings as the textual backbone with separately added enhancement components for multimodal integration. The current system was modular rather than end-to-end, combining a pre-trained language model with separate enhancement components. Modularity constrained the model's ability to uncover deeper and more complex interactions across text, audio, and visual modalities, and the framework design prevented full exploration of intricate cross-modal relationships, as acknowledged for future research.

Table 7 provides a consolidated summary of the reviewed methods, including their core approaches, techniques, key contributions, and identified limitations.

Table 8 provides a comparison of the four principal groups (Contextual-adaptive, Large Language Models, Lightweight, and Hybrid models), including the most used models in each category, their highlights, and their identified limitations.

Following the approaches discussed in the literature, a potential roadmap for sentiment analysis with transformers is illustrated in Fig. 10. The process begins with data collection and preprocessing, after which the choice of an appropriate transformer-based method must be made.

The first decision point concerns computational resource availability. If resources are not constrained, the next step is to determine whether sufficient labeled data is available. In cases where labeled data is limited, one must decide whether to extend the transformer with additional components. If no specialized modules (e.g., knowledge bases, knowledge graphs, or other deep learning-based components) are available to enhance the model's ability to capture complex dependencies, large language models (LLMs) and prompt engineering may be employed. Conversely, if adding such

Table 7 Comparison and summary of previous works

Article	Approach	Techniques	Highlights	Limitations
[17]	Contextual-Adaptive Models	RoBERTa, BERT, LDA, GB Classifier	Comparative evaluation of NLP models, including transformers on US Navy Twitter sentiment data; use of XAI and analysis of LDA impact on model performance	Lack of feature engineering parameter optimization, use of a single GB classifier for all models, and simplistic text preprocessing
[16]	Contextual-Adaptive Models	BERTimbau, m-BERT	Comparison of two BERT-based approaches on Brazilian Portuguese texts: pretrained with 13 vector aggregation methods and a logistic classifier, and fine-tuned with a linear layer and full weight updates	The dependency of the pretrained approach on a linear classifier, which poorly models complex nonlinear relationships
[90]	Contextual-Adaptive Models	BERT_BASE, SVR	Integration of BERT_BASE sentiment analysis with collaborative filtering and SVR in a hybrid recommender system	Domain limitation, data imbalance, and a need for improvement in the sentiment analysis component
[58]	Contextual-Adaptive Models	BERT SPC	Use of the BERT-SPC model for classifying sentence pairs combining embeddings and fully connected layers	Poor performance on the reliable evaluation of some datasets
[91]	Contextual-Adaptive Models	BERT, Supervised LDA	Proposal of an integrated framework using supervised LDA for fine-grained aspect extraction, auxiliary sentence construction based on semantic relations, and fine-tuning BERT for aspect-specific representation learning	Incorrect syntactic relations, wrong semantic candidate selection, and incomplete coverage in the absence of semantic candidates
[81]	Contextual-Adaptive Models	RoBERTa, BERTopic	Use of RoBERTa for sentiment analysis and BERTopic for topic modeling, leveraging BERT embeddings	Results are based only on a specific group of teachers and non-teachers, limiting generalization to the broader population
[92]	Contextual-Adaptive Models	RoBERTa, SiEBERT	Sentence tokenization with NLTK and application of two fine-tuned RoBERTa models (financial and SiEBERT) for sentiment evaluation of central bank speeches	Low data cleaning accuracy, limited country and data selection, without considering key factors like technological growth
[39]	Contextual-Adaptive Models	BERT, RoBERTa	Fine-tuning BERT and RoBERTa for multi-label sentiment classification of Guardian news articles with preprocessing and n-gram visualization	The use of social media data (SenWave), which includes abbreviations, internet slang, and emojis, complicated model training
[93]	Contextual-Adaptive Models	BERT-base-uncased, RoBERTa	Use of BERT and RoBERTa with translated and original data, combined with weak supervision auto-labeling via Snorkel	Snorkel does not support multilingual data, requiring prior translation before modeling

Table 7 (continued)

Article	Approach	Techniques	Highlights	Limitations
[47]	Contextual-Adaptive Models	mBERT, BanglaBERT, Bangla-Bert-Base, DistilmBERT, XLM-RoBERTa-base, Transformer-ensemble	Experiments on the Bangla review dataset using five transformer models and development of a Transformer-ensemble model to enhance accuracy	Lower performance on Bangla-English code-mixed data and difficulty in detecting the negative class due to data imbalance
[40]	Contextual-Adaptive Models	HybBERT, BERT, RoBERTa, DistilBERT	A combination of BERT, RoBERTa, and DistilBERT transformer models for simultaneous aspect extraction and sentiment classification	Unstable performance in some combinations, and evaluation is limited to a single dataset
[94]	Contextual-Adaptive Models	Spanish BERT, Hugging Face multilingual models, SVM, Naive Bayes, Logistic Regression	Training and comparison of Spanish and multilingual BERT models on Mexican COVID-19 Twitter data with standard evaluation metrics	Lexicon labeling errors, time and topic domain limitations, and exclusive focus on the Spanish language
[38]	Contextual-Adaptive Models	CT-BERT, BERT-base, BERT-large, DistilBERT, RoBERTa	Use of CT-BERT and Auxiliary Sentence approach in single- and pair-sentence scenarios for COVID-19 tweet sentiment analysis	Underperformance compared to general models due to non-English data and a lack of direct evaluation of the pair-sentence approach
[71]	Contextual-Adaptive Models	mBERT with shared-private architecture	Multilingual multi-task shared-private mBERT model for knowledge transfer between code-mixed sentiment analysis (main task) and English (auxiliary task)	Neglect of implicit sentiments and poor performance on noisy data and mixed sentiments
[95]	Contextual-Adaptive Models	Adapted Multimodal BERT (AMB), BERT	Use of a frozen BERT-based AMB model with adapter layers combining textual representations with audio-visual information in a layerwise FFN	No consideration of shift effects
[46]	Contextual-Adaptive Models	BERT, BERT_1, TM-BERT	Development of TM-BERT pretrained on a combination of Wikipedia and Twitter data and fine-tuned on the CV-SAT dataset for COVID-19 vaccination tweet sentiment analysis	Evaluation limited to COVID-19 Twitter data, dependence on data quality, and lack of testing generalizability to other domains or languages
[18]	Large Language Models	GPT-3, GPT-2, RuGPT-3	Use of GPT-3 with few-shot learning and GPT-2 fine-tuning for multilingual ASTE target triple extraction	Ineffectiveness of few-shot learning in smaller models like RuGPT-3 and the need for fine-tuning for adequate performance
[19]	Large Language Models	PaLM-540B	Semi-supervised learning with weakly labeled Reddit posts by PaLM-540B and knowledge distillation to smaller models using multiple predictions and chain-of-thought reasoning	Ethical considerations in model output use and the need for improved design and prompt optimization

Table 7 (continued)

Article	Approach	Techniques	Highlights	Limitations
[43]	Large Language Models	RoBERTa, LLaMA2	Comparison of RoBERTa (encoder-only) and LLaMA2 (decoder-only) in intent classification and sentiment analysis with fine-tuning, LoRA/QLoRA, few-shot, and zero-shot learning	Focus only on LLaMA2 decoder models, reliance on public data, and lack of input prompt optimization
[96]	Large Language Models	GPT-3.5, GPT-4, Llama 2, SiEBERT, RoBERTa	Empirical comparison of GPT-3.5, GPT-4, and Llama 2 with transformer learning models (SiEBERT, RoBERTa) in sentiment analysis over 20 + datasets, assessing data influence, prompting, and explainability	No fine-tuning, reduced performance on noisy data and multi-class tasks, and high dependency on prompt design
[44]	Large Language Models	ChatGPT-4	Use of collaborative platform VULL and ChatGPT-4 for user feedback sentiment analysis with direct prompting and manual result review	Challenges in understanding technical or cultural language and potential output biases
[97]	Large Language Models	BERT, GPT-2, LLaMA-2, Falcon, MistralAI	Quantification of social bias in five open-source LLMs via fine-tuning on social media data, employing 1500 simple and complex prompt tests and explainable methods for bias impact analysis	Use of formal prompts without sarcasm or toxic content, small sample size (100 per bias type), and need for deeper analysis of each bias category
[20]	Lightweight Models	DistilBERT, SVM + TF-IDF, NB + TF-IDF, CNN	Sentiment analysis using DistilBERT on the Rotten Tomatoes dataset with 20-token inputs and AdamW optimizer training	Small dataset size and input length limitation of 20 tokens, leading to information loss
[21]	Lightweight Models	BERTweet	Structured pruning of self-attention heads in BERTweet ranked by importance and architecture search for optimized tweet sentiment analysis	Evaluation only on BERTweet, and a high computational resource demand for method execution
[64]	Lightweight Models	DistilBERT	Use of DistilBERT with preprocessing, hyperparameter tuning, and class weighting for sentiment analysis on IMDB and CR datasets	Limited improvement over prior methods and low generalizability due to use of two small and simple datasets
[65]	Lightweight Models	BERT, Naive Bayes, SVM, Logistic Regression	Knowledge distillation from BERT to lightweight models (Naive Bayes, SVM, Logistic Regression) for IMDB sentiment analysis using soft labels	Mismatch of teacher-student feature space and reduction to 2000 samples, limiting linguistic diversity and increasing overfitting risk
[98]	Lightweight Models	CT-BERT, DistilBERT, Simple RNN, LSTM, GRU, Bi-RNN	Fine-tuning CT-BERT and comparison with Simple RNN, LSTM, GRU, BiRNN, and DistilBERT on the binary Toxic Comment dataset	Use of only a simple English binary dataset and problem simplification without examining data diversity or parameters

Table 7 (continued)

Article	Approach	Techniques	Highlights	Limitations
[99]	Hybrid Models	Cross-Mean Teacher Framework	Use of the Cross-Mean Teacher framework with student–teacher modules and random replacement data augmentation for knowledge transfer in cross-domain aspect-based sentiment analysis	High memory requirements for training two modules and low diversity in augmented samples
[100]	Hybrid Models	EAFL-ACSA	Introduction of the EAFL-ACSA mechanism for aspect-category sentiment analysis in federated learning, combining privacy preservation and data augmentation to improve representations	High computational resource needs, difficulty in pattern tuning and word representation, poor multi-class classification, and the possibility of incorrect polarity representation in random sentences
[101]	Hybrid Models	PGSO	Proposing two sequence optimization strategies (rule-based static and score-based dynamic) and the PGSO model with pattern construction and sequence tuner modules	Sometimes original sequence information is ignored, leading to incorrect predictions
[102]	Hybrid Models	SK-BERT, BERT	Integration of BERT with affective knowledge lexicons in the SK-BERT model to improve aspect-level sentiment classification using multi-head attention	Small and limited affective lexicons and model performance are dependent on lexicon quality and completeness
[103]	Hybrid Models	Loughran-McDonald Lexicon, BERT	Combining the Loughran-McDonald dictionary with BERT for financial text sentiment analysis, with fine-tuning after preprocessing	Model overfits after approximately 5 epochs, requiring early stopping or further regularization
[22]	Hybrid Models	Chinese BERT, BERT + FC, BERT + LSTM, BERT + CNN, TextCNN, TextRNN, Att-BLSTM, TextCRNN	Use of Chinese BERT with three architectures (BERT + FC, BERT + LSTM, BERT + CNN) for stock review sentiment analysis; BERT + FC selected as best model	Limited labeled data and evaluation restricted to a single dataset
[23]	Hybrid Models	Prompt-CSR, BERT	Use of a prompt-based BERT model with combined semantic refinement and multi-prompt learning to improve ABSA	Dependence on prompt quality and data, and high computational resources are required for automated prompt search
[104]	Hybrid Models	BERT-IAN	Use of BERT for semantic feature extraction, a transformer for deep feature extraction, and interactive attention between aspect and text	Lack of use of positional information

Table 7 (continued)

Article	Approach	Techniques	Highlights	Limitations
[105]	Hybrid Models	BERT-Invos	Use of BERT to encode text with multi-scale stacked dilated convolutions and nonlinear activation functions to improve feature extraction and context understanding	High computational complexity in pretraining and the stacked network
[106]	Hybrid Models	DAFS + GA-enhanced BERT with CNN	Combining dynamic attention-based feature selection (DAFS) with a genetic algorithm (GA) for feature and hyperparameter optimization and enhancing BERT with convolutional layers and aggregation	Relatively high running time
[107]	Hybrid Models	Adaptive Contextual Memory Graph Transformer (ACM-GT-ABSA)	Framework combining a dynamic knowledge graph, contextual memory, hierarchical graph encoder, contrastive learning, and semantic role labeling for aspect and sentiment extraction and analysis	Reduced adaptability in specialized domains, high computational complexity, and difficulty managing rapid emotional changes
[108]	Hybrid Models	CSMF-SPC, CSM-LRMFN, BiLSTM, BERT	Multimodal model (CSMF-SPC) with low-rank semantic fusion network (CSM-LRMFN) using BiLSTM for non-text modalities, BERT for text, and a self-adjusting loss function to improve polarity prediction	Model is effective only if all modalities (textual and non-textual) are fully available in the data
[74]	Hybrid Models	FinBERT, RNN, Bi-LSTM	Extraction of financial indicators with TA-lib, sentiment classification of news with FinBERT, integration of price data, and use of deep learning models (Bi-RNN, Bi-LSTM) for stock impact prediction	Focus is limited to English financial news and has reduced generalizability to other languages and domains
[42]	Hybrid Models	TF-BERT	Multimodal tensor-based model named TF-BERT with a TCF module added to BERT layers to enhance multimodal sentiment interaction and fusion	High computational cost limits efficiency in resource-constrained or real-time applications
[109]	Hybrid Models	CABiLSTM-BERT	Use of frozen BERT for feature extraction processed by CABiLSTM and a multi-head self-attention mechanism to preserve and integrate implicit features across layers and improve sentiment classification accuracy	Suboptimal performance in handling multi-level sentiment interactions and strong dependence on data balance

Table 7 (continued)

Article	Approach	Techniques	Highlights	Limitations
[110]	Hybrid Models	XLNet, BERT, RoBERTa, LightGBM	Integration of three powerful models, XLNet, BERT, and RoBERTa, in a hybrid architecture and use of LightGBM to exploit their outputs, with parameter fine-tuning and full BERT layer updates on the benchmark dataset	Scalability issues, lack of evaluation on larger datasets, complex real-world scenarios, and non-English languages
[111]	Hybrid Models	Word2Vec, GloVe, BERT-base, LSTM, GRU, Bi-GRU	Heterogeneous stacking framework with text preprocessing, three embedding types (Word2Vec, GloVe, BERT), and multi-stage classification using LSTM, GRU, and Bi-GRU at the base layer and LSTM at the meta layer	Weakness in extracting complex relationships compared to fully transformer-based models
[112]	Hybrid Models	BERT, BiLSTM, GCN, CNN	Combination of BERT for embeddings, BiLSTM for contextual feature extraction, GCN for syntactic dependency modeling, and CNN for feature compression	Lack of simultaneous consideration of all contextual factors and semantic relations; sensitivity to data quality and architecture tuning
[113]	Hybrid Models	BERT-CNN-BiLSTM	Integration of BERT for embeddings, CNN for local feature extraction, and BiLSTM for capturing bidirectional and long-term dependencies	High computational complexity and potential feature redundancy compared to BERT capabilities
[52]	Hybrid Models	AeBERT-BGRU	Combination of BERT with multi-head attention and bidirectional GRU for context-aware feature extraction and sentiment classification	High computational demand and strong dependency on preprocessing quality and domain adaptation
[114]	Hybrid Models	ESSGCN, BERT, GAT, GCN	Combination of aspect-based syntactic graph (with GAT) and semantic graph extracted from BERT (with GCN), with supervised and unsupervised contrastive learning to improve discriminability	Weakness in processing implicit sentiments and high dependency on syntactic structure quality and BERT capabilities
[45]	Hybrid Models	GloVe, Bi-GRU, 3D-CNN, COVAREP, Universal Transformer	Text feature extraction with GloVe and Bi-GRU, visual features with 3D-CNN, audio features with COVAREP, then fusion and processing with Universal Transformer	Shallow audio feature depth and partial loss of speech information
[115]	Hybrid Models	BERT-CBLBGA, BERT, TextCNN, BiLSTM, BiGRU	Use of BERT for embeddings, TextCNN for local features, BiLSTM and BiGRU for global features, combined with self-attention	High computational complexity and difficulty optimizing parameters
[116]	Hybrid Models	TopFuzz4SA, Transformer, FDNN	Combining the Transformer with a topic-aware attention mechanism and fuzzy deep neural network (FDNN) for extracting and refining textual features	Not designed or evaluated for dynamic and real-time sentiment analysis

Table 7 (continued)

Article	Approach	Techniques	Highlights	Limitations
[41]	Hybrid Models	Cross-modal Enhancement Network (CENet)	CENet model enhances textual representation by integrating visual and audio information in a transformer-based language model	Modular structure and lack of end-to-end learning of multimodal interactions

custom modules is feasible and beneficial, a hybrid transformer architecture can be adopted. If adequate labeled data is available, the model can instead be improved through fine-tuning of standard transformers, using context-adaptive modeling strategies. In scenarios where computational resources are limited, model compression techniques such as distillation or pruning are recommended to develop lightweight variants suitable for low-resource environments. Finally, the roadmap concludes with an evaluation of the chosen approach on a designated test dataset.

4.4 Evaluation of sentiment analysis methods

To address the fourth research question, “*Which metrics are used to evaluate transformer-based sentiment analysis models?*”, this section reviews the evaluation criteria reported in the surveyed studies. The literature indicates that the majority of works employ either classification-oriented metrics or regression-based measures to assess their proposed methodologies.

4.4.1 Classification-based metrics

Accuracy: Accuracy measures the proportion of correctly classified instances out of all instances in sentiment analysis. It is simple and widely used, but can be misleading in imbalanced datasets where one sentiment class dominates.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Precision: Precision indicates the fraction of predicted positive sentiments that are actually positive. It is valuable when minimizing false positives is important, such as avoiding mislabeling neutral text as negative.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

Recall: Recall measures the fraction of actual positive sentiments that the model correctly identifies. It is crucial to

capture all positive matters, like detecting all negative feedback.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

F1-Score: The F1-Score is the harmonic mean of precision and recall, balancing both metrics. It is preferred in imbalanced sentiment tasks as it penalizes models with large disparities between precision and recall.

$$\text{F1 - Score} = 2 \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (4)$$

Macro-F1: Macro-F1 averages the F1-score of each sentiment class equally, regardless of class size. It is ideal for imbalanced datasets, ensuring minority classes like neutral sentiments are fairly evaluated.

$$\text{Macro - F1} = \frac{1}{C} \sum_{i=1}^C F1_i \quad (5)$$

Micro-F1: Micro-F1 computes a global F1 by aggregating counts across all classes first. It equals accuracy in single-label tasks but favors larger classes in imbalanced sentiment data.

$$\text{micro}_{\text{recall}} = \frac{TP}{TP + FN} \quad (6)$$

$$\text{micro}_{\text{precision}} = \frac{TP}{TP + FP} \quad (7)$$

$$\text{Micro - F1} = 2 \cdot \frac{\text{micro}_{\text{precision}} \cdot \text{micro}_{\text{recall}}}{(\text{micro}_{\text{precision}} + \text{micro}_{\text{recall}})} \quad (8)$$

Weighted-F1: Weighted-F1 averages per-class F1 scores weighted by the number of samples in each class. It handles imbalance by emphasizing performance on more frequent sentiments while still considering others.

$$\text{Weighted - F1} = \sum_{i=1}^C w_i \cdot F1_i \quad (9)$$

Cohen's Kappa: Cohen's Kappa measures agreement between predictions and true labels, adjusted for chance. It

Table 8 Comparison of four categories of transformer-based sentiment analysis

Approaches	Short description	Models	Highlights	Limitations
Contextual-adaptive models	Contextual-adaptive models leverage pretrained transformers (e.g., BERT, RoBERTa) and adapt them through various techniques like domain-specific pretraining and fine-tuning to address the complexities of social media text. These models extend adaptability to linguistic contexts, particularly through multilingual transformers	BERT RoBERTa Domain-specific adaptations (e.g. FinBERT)	Tailored for challenges such as short, noisy, and informal texts Outperform traditional baselines by significant margins Achieve higher accuracy and robustness in sentiment analysis tasks on challenging datasets	Poor generalization outside the target domain/language; requires large labeled (or unlabeled) domain corpora expensive and time-consuming retraining/fine-tuning for each new domain Difficulty in detecting implicit sentiments
Large language models (LLMs)	LLMs represent a shift towards decoder-only generative transformer architectures that perform various sentiment-related tasks through in-context learning and prompt engineering, often with minimal task-specific data aiming to eliminate or drastically reduce fine-tuning and labeled data needs	GPT-3, GPT-4 LLaMA-2 PaLM-540B	Potential to reduce the need for large annotated datasets and extensive fine-tuning Capable of performing complex sentiment tasks, from polarity classification to intent detection Can be steered towards high performance using prompt engineering	Performance is highly dependent on prompt design (e.g., choice of role, output format) Significant performance degradation when applied to noisy, informal social media texts Risk of hallucinations and bias
Lightweight models	Lightweight models prioritize speed and efficiency for real-world deployment, using techniques like knowledge distillation and pruning to create smaller, faster models that maintain reasonable performance	DistilBERT (most common) quantized BERT variants	DistilBERT mostly outperforms traditional machine learning baselines and older deep learning models Reduces memory footprint and inference latency, making it suitable for resource-constrained devices	Noticeable accuracy drop on complex/nuanced cases; fragile efficiency–accuracy trade-off; evaluations often limited to a few datasets; aggressive truncation harms long-context understanding; distillation itself is expensive
Hybrid models	Hybrid models enhance and combine BERT-family models with additional mechanisms to improve performance on difficult sentiment analysis tasks	BERT combined with CNNs, RNNs (e.g., BiLSTM, BiGRU), GCNs, and other architectures Use of external sentiment lexicons and multimodal fusion modules	Aim to push performance higher on challenging sentiment tasks Incorporate various innovative mechanisms to address remaining accuracy gaps	High computational complexity and memory demands Long training times and reliance on the quality of external components (e.g., lexicons, dependency parsers) Complexity may complicate deployment and scalability in real-world applications

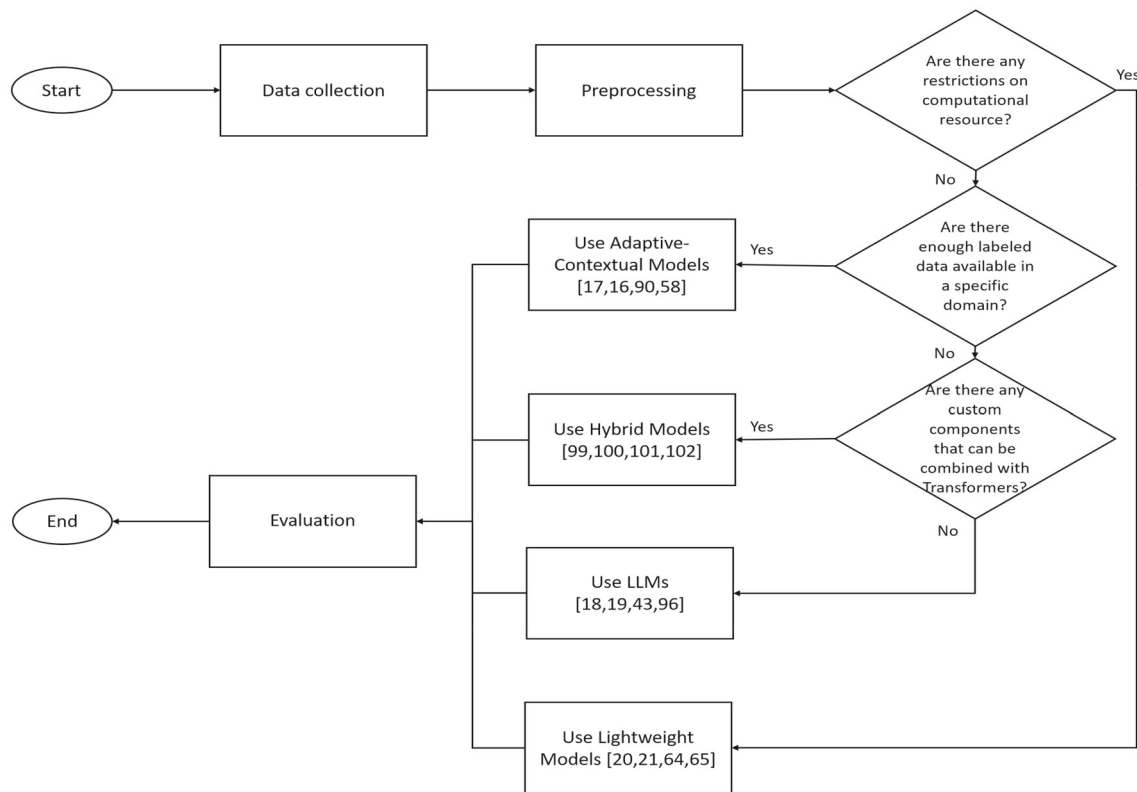


Fig. 10 The roadmap for sentiment analysis using transformers

provides a more robust evaluation than accuracy in sentiment tasks with potential random agreement.

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (10)$$

Matthews Correlation Coefficient (MCC): MCC assesses binary classification quality using all confusion matrix elements in a balanced way. It is robust to imbalance and preferred for binary sentiment tasks like positive vs. negative.

$$\text{Mathew Correlation Coefficient} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (11)$$

Acc-2/Acc-7: Acc-2 and Acc-7 are accuracy measures for 2-class or 7-class fine-grained sentiment tasks. They evaluate correct predictions across intensity levels, which is useful in nuanced sentiment analysis.

$$\text{Accuracy} = \frac{\sum_{i=1}^C TP_i}{N} \quad (12)$$

Specificity: Specificity measures the proportion of actual negative instances correctly identified as negative. It is important in binary sentiment tasks for accurately handling the negative or neutral class.

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (13)$$

AUC (Area Under the Curve): AUC generally refers to the area under a performance curve (most commonly ROC in classification), quantifying a model's discriminative power between classes. In sentiment analysis, it is often used interchangeably with ROC-AUC for binary or multi-class tasks, providing a single scalar summary where higher values (up to 1) reflect stronger separation of sentiment categories.

$$\text{TPR} = \frac{TP}{TP + FN} \quad (14)$$

$$\text{FPR} = \frac{FP}{FP + TN} \quad (15)$$

ROC-AUC: ROC-AUC summarizes the model's ability to distinguish classes across thresholds. Values range from 0.5 (random) to 1 (perfect), making it threshold-independent for comparison.

4.4.2 Regression-based metrics

Mean Absolute Error (MAE): MAE calculates the average absolute difference between predicted and actual sentiment scores. It is intuitive for numerical predictions, like 1–5 star ratings, with lower values indicating better accuracy.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (16)$$

Root Mean Squared Error (RMSE): RMSE is the square root of the average squared differences, penalizing larger errors more. It measures prediction error in the same units as sentiment scores, sensitive to outliers.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (16)$$

Pearson Correlation Coefficient: Pearson correlation quantifies the linear relationship between predicted and actual sentiment values. It ranges from -1 (strong negative) to $+1$ (strong positive), assessing alignment in variations.

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}} \quad (18)$$

Table 9 summarizes the employed evaluation metrics in the paper. This table provides a brief definition for each metric, publications that have used this metric, highlights the advantages, and describes its use cases.

Our review critically assesses the adequacy and comprehensiveness of these metrics across different tasks:

- **Handling Data Imbalance:** We explicitly noted in Sect. 4.4.1 that while Accuracy is a common benchmark, it is inadequate for imbalanced datasets. Consequently, we highlighted the use of Macro-F1 and Weighted-F1 as more comprehensive measures that prevent bias toward majority classes.
- **Fine-Grained & Multimodal Adequacy:** For tasks requiring nuanced sentiment scales (e.g., -3 to $+3$ in Multimodal analysis), simple binary metrics are insufficient. We discussed the adoption of Acc-7 (7-class accuracy) and Pearson Correlation to adequately capture sentiment intensity and alignment between predicted and actual values.
- **Robustness:** In Sect. 7 (Conclusion), we emphasize that no single metric is fully comprehensive. Therefore, we advocate for using a combination of classification and regression metrics (e.g., F1-score alongside MAE) to ensure a robust and holistic evaluation of transformer-based models.

4.5 Quality assessment

The primary motivation for conducting a quality assessment of the reviewed articles is to address the main research question of our systematic literature review (SLR), namely RQ5:

- **RQ5:** *How has the maturity and scientific quality of transformer-based sentiment analysis research evolved in recent years?*

To provide a rigorous and non-descriptive answer to this question, we designed four Quality Assessment Questions (QAQs) aimed at evaluating and quantifying the contributions of articles from multiple perspectives.

With these four questions, a wide range of aspects has been covered. QAQ1 evaluates the scientific and technical soundness, ensuring the method rests on a solid foundation with appropriate theoretical principles, mathematical modeling, and valid assumptions. QAQ2 focuses on the clarity and reproducibility of the presentation, checking whether the method's steps, algorithms, and procedures are described in sufficient detail and logically to enable understanding and replication by others. QAQ3 examines the thoroughness of the evaluation, verifying that the method undergoes rigorous testing through appropriate experiments, benchmarks, comparisons to existing work, meaningful metrics, statistical analysis, and honest discussion of limitations. Finally, QAQ4 assesses the credibility of the sources, considering the authors' expertise, the paper's integrity, and the reputation of the publishing journal or conference to guard against low-quality or predatory publications.

The quality of each study is then converted into quantitative scores based on predefined criteria. These scores are subsequently compared across different years, and a trend line is constructed to illustrate the progression of sentiment analysis methods in recent years.

- **QAQ1:** Does the proposed method demonstrate a solid scientific/technical foundation and an appropriate mathematical model?
- **QAQ2:** Are the steps of the proposed method presented and explained with sufficient clarity?
- **QAQ3:** Does the proposed method receive a comprehensive evaluation?
- **QAQ4:** Do the authors, the article itself, and the publishing journal possess academic credibility and recognized value?

Then, a score is assigned to each article for each QAQ, as

Table 9 Major evaluation metrics for sentiment analysis using transformers

Metric	References	Definition	Advantages	Contexts of use
Accuracy	[45, 113, 117]	The ratio of correctly predicted instances to the total instances	Simple to understand; useful for balanced datasets	General classification tasks; balanced classes
Precision	[40, 111, 113, 167]	The ratio of true positive predictions to the sum of true positives and false positives	Indicates the quality of positive predictions; useful in scenarios where false positives are costly	Imbalanced datasets; when false positives are critical
Recall	[75, 76, 117, 167]	The ratio of true positive predictions to the sum of true positives and false negatives	Measures the model's ability to identify all relevant instances; important in critical applications	Imbalanced datasets; when false negatives are critical
F1-Score	[40, 47, 80, 118, 167]	The harmonic mean of precision and recall balances both metrics	Provides a single score to evaluate the balance between precision and recall	Imbalanced datasets; when both false positives and negatives matter
Macro-F1	[23, 118, 119, 168, 169]	The average F1-score across all classes, treating all classes equally	Highlights performance across all classes; not influenced by class imbalance	Multi-class classification tasks
Micro-F1	[43, 54, 99, 170]	The F1-score is calculated globally across all classes by counting the total true positives, false negatives, and false positives	More sensitive to class imbalance; focuses on overall performance	Multi-class classification tasks, especially with imbalanced data
Weighted F1	[103, 123, 166, 171, 172]	The F1-score is calculated for each class and weighted by the number of true instances for each class	Accounts for class imbalance; provides a more representative score for imbalanced datasets	Multi-class classification tasks with significant class imbalance
Cohen's Kappa	[47, 60, 173–175]	A statistic that measures inter-rater agreement for categorical items; it compares observed accuracy to expected accuracy	Adjusts for chance agreement; useful for evaluating model performance beyond accuracy	Classification tasks, especially with multiple annotators
MCC (Matthews Correlation Coefficient)	[17, 90, 176, 177]	A measure of the quality of binary classifications, taking into account true and false positives and negatives	A balanced measure that can be used even if the classes are of very different sizes	Binary classification tasks; imbalanced datasets
Acc-2	[41, 123, 148, 150, 152]	Accuracy of the top 2 predictions in multi-class classification tasks	Useful for evaluating models where the top predictions are important (e.g., recommendation systems)	Multi-class classification tasks
Acc-7	[42, 61, 108, 123, 152]	Accuracy of the top 7 predictions in multi-class classification tasks	Similar to Acc-2, it allows for flexibility in classification tasks where multiple correct answers are possible	Multi-class classification tasks

Table 9 (continued)

Metric	References	Definition	Advantages	Contexts of use
Specificity	[75, 178]	The ratio of true negatives to the sum of true negatives and false positives measures the proportion of actual negatives that are correctly identified	Useful for understanding how well the model identifies negative cases; complements recall	Medical diagnostics: when false positives are critical
AUC (Area Under the Curve)	[90, 107, 179]	The area under the ROC curve represents the model's ability to discriminate between positive and negative classes	Provides a single score to evaluate model performance across all classification thresholds	Binary classification tasks: evaluating model discrimination ability
ROC-AUC	[76, 82, 98, 110, 180]	The Receiver Operating Characteristic curve plots the true positive rate against the false positive rate	Visual representation of model performance; AUC provides an aggregate measure of performance	Binary classification tasks; model comparison
MAE (Mean Absolute Error)	[41, 73, 107, 123, 181]	The average of absolute differences between predicted and actual values	Simple to interpret; provides a direct measure of prediction accuracy in regression tasks	Regression tasks
RMSE (Root Mean Squared Error)	[74, 90]	The square root of the average of squared differences between predicted and actual values	Sensitive to outliers; provides a measure of prediction accuracy that penalizes larger errors more heavily	Regression tasks
Pearson Correlation Coefficient	[61, 63, 108, 152, 153]	A measure of the linear correlation between two variables, ranging from -1 to 1	Indicates the strength and direction of a linear relationship; useful for assessing model performance in regression	Regression tasks: evaluating relationships between predicted and actual values

follows:

- **QAQ1**
 - o **Y (Yes):** The proposed method is supported by a strong scientific/technical foundation, as well as a suitable mathematical model.
 - o **P (Partial):** The proposed method satisfies one of these conditions (scientific/technical foundation or mathematical model).
 - o **N (No):** The proposed method lacks both.
 - o *Note:* A strong scientific/technical foundation implies sound assumptions, a clearly defined research roadmap, a coherent workflow, and validation through credible sources. A suitable mathematical model refers to the use of established, logically consistent, and verifiable mathematical formulations.
- **QAQ2**
 - o **Y (Yes):** The method has a transparent workflow, well-defined pseudocode, and a comprehensive discussion of all components.
 - o **P (Partial):** The method lacks clarity and cannot be reliably replicated.
 - o **N (No):** The method remains too abstract.
- **QAQ3**
 - o **Y (Yes):** The study includes both a systematic performance analysis and an ablation study.
 - o **P (Partial):** The study provides only a systematic performance analysis.
 - o **N (No):** The evaluation is either absent or deferred to future work.
- **QAQ4**
 - o **Y (Yes):** Assigned if the normalized journal value lies within [1, 0.5).
 - o **P (Partial):** Assigned if the value lies within [0.5, 0).
 - o **N (No):** Assigned if the value equals 0.

Journal Value: Since the articles included in this SLR are published in journals with varying impact factors, it is necessary to normalize these values to the interval [0, 1]. To achieve this, the min–max normalization technique is applied, as expressed in Eq. (19). In this equation, IF_i represents the impact factor of journal i , $\min(IF)$ denotes the lowest impact factor among the journals considered in this SLR, $\max(IF)$ indicates the highest impact factor, and $Journalvalue_i$ corresponds to the normalized value assigned to journal i .

$$Journalvalue_i = \frac{IF_i - \min(IF)}{\max(IF) - \min(IF)} \quad (19)$$

The scoring scheme assigns a value of **1** for *Yes (Y)*, **0.5** for *Partial (P)*, and **0** for *No (N)*. Table 10 reports the scores for each article.

Drawing on the quality assessment results presented in Table 10 and Fig. 11, it is evident that research on transformer-based sentiment analysis has advanced considerably in recent years. Specifically, the quality of studies in this domain has shown a consistent upward trend since 2020.

Figure 11 illustrates the average scores obtained for each of the four QAQs during the period 2020–2025. The average QAQ scores are also aggregated over two-year intervals, and a trend line is plotted to highlight changes over time. The upward trajectory of this line clearly reflects the steady growth in the average annual quality scores of published articles.

Accordingly, in response to Q5, the evidence demonstrates that the quality of articles published between 2020 and 2025 has not only improved but continues to rise. Furthermore, it is noteworthy that large language models (LLMs) have played a pivotal role in advancing both the maturity and scientific quality of this body of research.

5 Discussions

This chapter synthesizes the main findings of the systematic literature review (SLR) and reflects on their research implications and directions for future work. Drawing on the analysis of existing approaches, evaluation methodologies, and the challenges identified in this review, we highlight key trends and major themes for further discussion.

The study of critical challenges in Chapter 2 served as an entry point for considering the future trend of transformer-based sentiment analysis. Although those challenges reveal concrete research gaps, the future scope of this field extends well beyond them. Accordingly, this chapter not only addresses the issues previously raised but also explores emerging trends, cross-disciplinary opportunities, and new paradigms that are likely to shape the advancement of this research domain.

To this end, the future directions Sect. 5.1 outlines potential research pathways and evolving trends in transformer-based sentiment analysis. Following this, the Synthesis Sect. 5.2 consolidates the fundamental insights related to the SLR's research questions and proposes a framework to guide the development of future studies in this area.

5.1 Future directions

Addressing Linguistic Bias: One of the most prominent limitations in current research lies in its strong dependence on English-language datasets. Addressing this linguistic bias requires the development of robust multilingual sentiment analysis frameworks. This challenge extends well beyond straightforward translation, as it must account for cultural nuances, code-mixing, and the specific complexities of low-resource languages. Future work should therefore prioritize the construction of comprehensive multilingual datasets with balanced representation across languages and cultural contexts. Advancements in cross-lingual transfer learning will be especially vital for languages with limited labeled resources. Achieving meaningful progress in this area will necessitate close collaboration between computational linguists and cultural experts to ensure that sentiment is captured accurately across diverse linguistic communities. Additionally, frameworks such as Snorkel, which currently depend on translation before processing, represent a bottleneck that future research should overcome through native multilingual capabilities.

High Computational Costs and Lightweight Solutions: Another major challenge is the trade-off between model performance and computational efficiency. Many existing approaches rely on hybrid architectures that combine transformers with traditional machine learning components, thereby constraining their ability to capture complex relationships. Future research should move toward fully transformer-based, end-to-end systems capable of eliminating architectural inconsistencies while remaining computationally feasible. The high computational cost of current methods poses a significant barrier to real-world deployment, particularly in resource-constrained settings. This issue becomes even more pronounced in mobile and edge computing environments. To address this, researchers should explore lighter transformer variants and efficient pruning strategies that reduce overhead without sacrificing accuracy. Developing adaptive systems that can dynamically adjust their complexity based on input characteristics also offers a promising avenue for expanding accessibility across diverse computational environments.

Data Imbalance and Noise Handling Strategies: The persistent challenge of imbalanced datasets, especially concerning minority sentiment classes, calls for innovative strategies beyond conventional resampling. Future studies should investigate advanced data augmentation methods that

Table 10 Quality assessment of articles based on QAQs

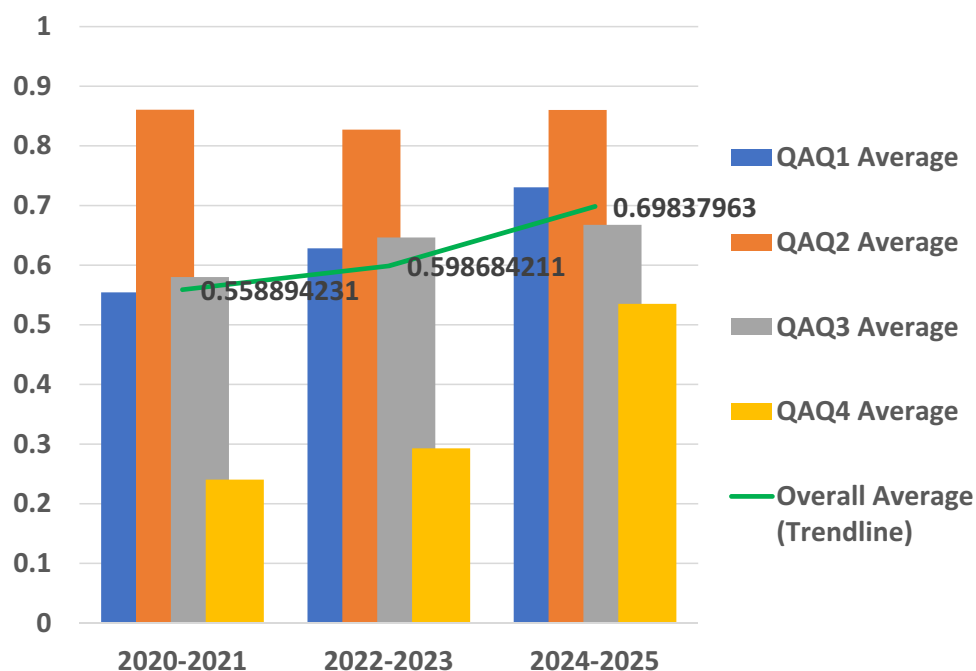
References	Year	QAQ1	QAQ2	QAQ3	QAQ4	Total score	Average QAQ1-QAQ4 (for each year)
[98]	2020	N	Y	P	N	1.5	0.531
[104]	2020	N	Y	P	N	1.5	
[117]	2020	P	Y	P	N	2	
[182]	2020	P	P	P	N	1.5	
[183]	2020	N	P	Y	N	1.5	
[67]	2020	P	P	P	N	1.5	
[120]	2020	Y	Y	P	N	2.5	
[57]	2020	P	Y	Y	Y	3.5	
[184]	2020	N	Y	P	N	1.5	
[185]	2020	P	Y	P	Y	3	
[186]	2020	Y	Y	P	P	3	0.586
[187]	2020	P	Y	P	P	2.5	
[17]	2021	Y	Y	P	Y	3.5	
[22]	2021	P	P	Y	P	2.5	
[38]	2021	Y	Y	P	N	2.5	
[188]	2021	P	P	P	N	1.5	
[189]	2021	P	P	P	N	1.5	
[190]	2021	P	P	P	P	2	
[118]	2021	P	Y	P	N	2	
[191]	2021	P	Y	P	N	2	
[72]	2021	Y	Y	Y	N	3	0.572
[192]	2021	Y	Y	Y	N	3	
[193]	2021	Y	Y	P	N	2.5	
[194]	2021	Y	Y	P	P	3	
[119]	2021	N	Y	N	P	2	
[45]	2022	N	Y	P	N	1.5	
[112]	2022	Y	Y	Y	P	3.5	
[65]	2022	P	P	P	N	1.5	
[102]	2022	Y	Y	P	N	2.5	
[111]	2022	P	Y	P	Y	3	
[116]	2022	Y	Y	Y	P	3.5	0.572
[52]	2022	P	Y	P	P	2.5	
[16]	2022	P	P	P	N	1.5	
[46]	2022	P	Y	P	N	2	
[195]	2022	P	P	P	N	1.5	
[196]	2022	Y	P	P	N	2	
[169]	2022	P	Y	Y	N	2.5	
[147]	2022	N	Y	Y	P	2.5	
[55]	2022	Y	P	P	N	2	
[197]	2022	Y	Y	P	P	3	
[115]	2022	P	Y	P	P	2.5	0.572
[198]	2022	Y	Y	P	N	2.5	
[199]	2022	N	P	P	N	1	

Table 10 (continued)

References	Year	QAQ1	QAQ2	QAQ3	QAQ4	Total score	Average QAQ1-QAQ4 (for each year)
[200]	2022	Y	Y	P	P	3	0.625
[40]	2023	Y	Y	P	N	2.5	
[20]	2023	N	Y	P	N	1.5	
[115]	2023	P	Y	Y	Y	3.5	
[93]	2023	P	Y	P	N	2	
[18]	2023	P	P	P	N	1.5	
[19]	2023	P	P	Y	N	2	
[23]	2023	Y	Y	Y	Y	4	
[95]	2023	N	P	Y	N	1.5	
[113]	2023	N	P	P	N	1	
[94]	2023	P	P	P	P	2	
[64]	2023	P	P	P	N	1.5	
[41]	2023	Y	Y	Y	Y	4	
[201]	2023	P	P	P	N	1.5	
[202]	2023	P	P	P	N	1.5	
[203]	2023	Y	Y	P	N	2.5	
[204]	2023	P	P	P	N	1.5	
[122]	2023	Y	Y	P	N	2.5	
[121]	2023	Y	Y	P	N	2.5	
[145]	2023	Y	Y	Y	Y	4	
[205]	2023	P	Y	P	P	2.5	
[206]	2023	Y	Y	Y	Y	4	
[207]	2023	Y	Y	Y	Y	4	
[161]	2023	N	Y	P	Y	2.5	0.717
[148]	2023	Y	Y	Y	Y	4	
[90]	2024	Y	Y	P	N	2.5	
[21]	2024	Y	Y	P	Y	3.5	
[99]	2024	Y	Y	Y	P	3.5	
[58]	2024	Y	Y	P	N	2.5	
[105]	2024	Y	Y	Y	P	3.5	
[91]	2024	P	Y	P	Y	3	
[81]	2024	Y	P	N	P	2	
[108]	2024	P	Y	Y	P	3	
[74]	2024	Y	Y	P	N	2.5	
[100]	2024	Y	Y	Y	Y	4	
[92]	2024	N	P	N	N	0.5	
[47]	2024	N	Y	P	N	1.5	
[96]	2024	N	Y	P	N	1.5	
[71]	2024	Y	Y	Y	P	3.5	
[114]	2024	Y	Y	P	Y	3.5	
[140]	2024	P	N	P	N	1	
[66]	2024	N	P	P	Y	2	
[208]	2024	Y	P	P	N	2	

Table 10 (continued)

References	Year	QAQ1	QAQ2	QAQ3	QAQ4	Total score	Average QAQ1-QAQ4 (for each year)
[171]	2024	Y	P	P	N	2	0.679
[209]	2024	Y	Y	Y	Y	4	
[139]	2024	Y	Y	Y	Y	4	
[53]	2024	P	Y	Y	N	2.5	
[75]	2024	Y	Y	Y	Y	4	
[156]	2024	Y	Y	Y	Y	4	
[128]	2024	Y	Y	Y	Y	4	
[210]	2024	Y	Y	P	I	3.5	
[124]	2024	Y	Y	Y	Y	4	
[44]	2025	N	P	N	Y	1.5	
[107]	2025	P	Y	Y	Y	3.5	
[109]	2025	P	Y	Y	Y	3.5	
[110]	2025	Y	Y	P	P	3	
[103]	2025	P	Y	P	N	2	
[97]	2025	N	P	N	Y	1.5	
[39]	2025	Y	Y	P	Y	3.5	
[101]	2025	Y	Y	Y	Y	4	
[106]	2025	Y	Y	P	P	3	
[42]	2025	Y	Y	Y	Y	4	
[172]	2025	P	N	P	N	1	
[70]	2025	P	P	P	P	2	
[211]	2025	Y	Y	P	P	3	
[142]	2025	Y	Y	Y	P	3.5	
[77]	2025	N	P	P	P	1.5	
[138]	2025	P	Y	Y	Y	3.5	
[212]	2025	Y	Y	Y	N	3	
[83]	2025	Y	Y	P	N	2.5	
[80]	2025	Y	Y	P	N	2.5	
[69]	2025	P	Y	P	N	2	
[213]	2025	Y	Y	P	P	3	
[68]	2025	P	Y	P	Y	3	
[214]	2025	Y	Y	Y	P	3.5	
[163]	2025	P	P	P	N	1.5	
[43]	2025	P	P	P	P	2	
[134]	2025	P	P	P	N	1.5	
[130]	2025	P	Y	Y	Y	3.5	
[152]	2025	P	Y	P	N	2	
[215]	2025	Y	Y	Y	Y	4	
[127]	2025	Y	Y	Y	P	3.5	

Fig. 11 Progress in sentiment analysis using transformers

move past simple random replacement and instead harness generative models to produce synthetic yet realistic training samples. This issue becomes more complex in the context of noisy social media data, where informal language, abbreviations, internet slang, and emojis introduce further parsing difficulties. Designing specialized preprocessing pipelines that preserve semantic meaning while handling such informal features represents an important research direction. Furthermore, constructing larger and more diverse datasets that span multiple domains, time periods, and cultural contexts will provide a stronger foundation for developing more resilient models.

Implicit Sentiment and Nuance Detection: Current models continue to struggle with implicit sentiment and subtle contextual cues. Their inability to accurately detect irony and sarcasm, often overlooked by lexicon-based methods, remains a critical limitation. Progress in this area will require advanced techniques capable of capturing implicit sentiment without explicit surface markers, drawing on deeper insights from linguistic pragmatics and contextual inference. Future work should also explore methods for detecting mixed emotions and emotional shifts within a single text, as real-world communication frequently conveys complex affective states that current binary or multiclass classification frameworks fail to capture. Enhancing models' ability to account for cultural and domain-specific contexts will be equally crucial for ensuring the successful deployment of sentiment analysis systems across a wide range of applications.

Real-Time and Streaming Analysis: The current inability of most models to perform real-time sentiment analysis opens up important opportunities for future research. Building systems that can process instant messages and live conversations will require fundamental advances in streaming architectures. Future studies should focus on dynamic sentiment classification methods that can adapt to rapidly shifting emotional contexts during ongoing interactions. Developing streaming frameworks optimized for continuous monitoring would enable practical applications in areas such as social media surveillance, automated customer service, and real-time market sentiment tracking. However, realizing this vision will require overcoming computational challenges to ensure high accuracy while handling continuous data streams.

Interpretability and Explainable AI (XAI): Another critical issue lies in the black-box nature of transformer-based models, which makes it difficult to interpret their decision-making processes, especially in domains where transparency and accountability are essential. Future research should therefore prioritize the development of interpretable techniques, such as attention visualizations that clearly highlight the text elements driving sentiment predictions. Researchers should move beyond simplistic attention weights and explore feature importance methods tailored to transformer architectures. In addition, building natural-language explanation systems that provide human-understandable justifications for model outputs would represent an important step toward

enhancing user trust and enabling deployment in sensitive contexts. Likewise, counterfactual explanation methods, which reveal how altering specific input elements changes predictions, could offer valuable insights into both model behavior and potential biases.

Robust Multimodal Fusion and Missing Modalities:

In the area of multimodal sentiment analysis, current models struggle to handle missing modalities effectively, creating further opportunities for innovation. Future work should develop advanced feature-fusion strategies capable of robustly integrating textual, acoustic, and visual information, while also remaining resilient to incomplete data. Progress will also depend on exploring cross-modal attention mechanisms that can more effectively capture interactions between modalities. A particularly promising direction involves unified representation learning approaches, which aim to integrate information across modalities seamlessly while retaining interpretability. This research is especially relevant to social media analytics, where posts often combine text with images or videos.

Ethical deployment and social bias mitigation: Future research should focus on the responsible deployment of sentiment analysis systems to prevent the reinforcement of harmful stereotypes or biases. Developing comprehensive frameworks for detecting and mitigating social bias will require interdisciplinary collaboration among computer scientists, social scientists, and ethicists. In addition, the creation of transparency frameworks that render model decision-making processes auditable and accountable is vital for fostering public trust and ensuring ethical deployment.

Advanced Evaluation and Reproducibility Frameworks: The limitations of current evaluation methodologies highlight the need for more advanced metrics that extend beyond conventional accuracy measures. Future studies should examine human–AI collaborative evaluation frameworks, which can provide more nuanced assessments of model performance. Likewise, the establishment of standardized benchmarks across diverse architectures and approaches would enable more meaningful cross-study comparisons. Further, advancing reproducibility frameworks that account for sensitivity to hyperparameter variations and prompt formulations will be crucial for ensuring reliable and trustworthy scientific progress in this field.

Long-Text Processing and Longitudinal Analysis:

Expanding sentiment analysis into specialized domains presents both opportunities and challenges. Future research should investigate techniques for processing long texts that exceed current token limitations, such as through hierarchical approaches. Moreover, developing models capable of tracking sentiment evolution over time and capturing patterns of persistence will open new possibilities for longitudinal analysis and trend forecasting.

Prompt Optimization and Efficient Learning

Paradigms: Finally, the limitations of smaller, locally deployable models in few-shot learning scenarios create opportunities for developing more efficient learning paradigms. Promising directions include automated prompt optimization techniques that extend and improve human-crafted prompts, as well as domain-specific prompt injection methods. Collaborative strategies that combine human annotators with large language models (LLMs) could also enhance task performance while reducing labeling costs. In addition, research into meta-learning approaches, capable of adapting rapidly to new domains or languages with minimal training data, could significantly accelerate the practical adoption of sentiment analysis systems.

Challenges in Specialized Domains and Contextual

Ambiguity: While general-domain transformer models demonstrate remarkable capabilities, their direct application to specialized high-stakes fields—such as finance, healthcare, and law—remains a critical bottleneck. A significant gap identified in our review is the inability of general PLMs to handle domain-specific jargon and contextual ambiguity. As highlighted in recent studies on financial sentiment analysis, standard pre-trained models often fail to grasp nuanced semantic shifts where vocabulary carries distinct professional implications (e.g., "liability" or "bullish" in market reports versus general conversation). Similarly, in the medical domain, general BERT models struggle with clinical terminology and specific contexts like vaccination discourse without domain-specific pre-training.

Data Constraints and Privacy in Sensitive Sectors: The curation of high-quality datasets for niche fields is severely hindered by privacy concerns and the scarcity of expert annotators. In sectors like decentralized finance (DeFi) and cybersecurity, there is a distinct lack of labeled benchmarks

for tasks such as scam detection or intrusion sensing. Furthermore, relying on public datasets creates privacy risks in sensitive sectors like healthcare and banking. To address this, future research must explore privacy-preserving sentiment analysis frameworks. Techniques such as Federated Learning, which allow models to learn from decentralized data without exposing sensitive user information, represent a vital but under-explored direction for these sensitive industries.

Tailored Research Opportunities: To overcome the limitations of data scarcity and domain mismatch, future research should prioritize three key areas. First, instead of relying solely on standard fine-tuning, researchers should investigate domain-adaptive pretraining strategies—such as the AdaFT framework proposed for financial texts—that inject domain-specific knowledge into the model architecture before the classification stage. Furthermore, given the high cost of annotation in fields like law, developing robust zero-shot and few-shot learning frameworks using advanced prompting strategies is essential; studies suggest that specialized Large Language Models (e.g., FinGPT or BloombergGPT) combined with optimized prompt engineering can significantly improve performance in low-resource settings. Finally, future work must focus on robust cross-domain generalization strategies that can align feature spaces between resource-rich source domains and resource-poor target domains, for instance, by transferring sentiment knowledge from general social media to specialized urban planning feedback.

Multilingual and Cross-Lingual Adaptation Challenges: A critical gap identified in current literature is the limited capability of models to handle diverse linguistic contexts beyond English. While data scarcity is a factor, the challenge extends to the architectural inability to handle specific linguistic intricacies such as morphological complexity (common in agglutinative languages) and code-switching (e.g., mixing languages in social media). Current translation-based approaches often fail to capture these nuances. Future research must prioritize the adoption and refinement of multilingual transformer architectures (such as mBERT, XLM-R, and IndicBERT) that offer native cross-lingual understanding. Developing frameworks capable of "zero-shot cross-lingual transfer" will be essential for applying sentiment analysis to low-resource languages without relying on massive labeled datasets.

5.2 Synthesis

This study proposes a practical development guide for sentiment analysis, building on the discussions presented in the preceding sections. In this section, we identify common trends, strengths, and limitations within that category, and highlight gaps or open problems in that specific category and the recommended approach for each type of sentiment analysis, which has been introduced in Sect. 2.1. Subsequently, we outline approaches to sentiment analysis according to its different types. Moreover, we describe the appropriate methods and evaluation criteria applicable to each category. Figure 12 presents a schematic overview of this guide, encompassing the major types of sentiment analysis addressed in this review, including polarity-based analysis, aspect-based analysis, intention-oriented analysis, fine-grained sentiment analysis, subjectivity/objectivity classification, and multilingual sentiment analysis. For each type, we suggest the most suitable analytical approach, evaluation methods, and commonly used metrics. It should be emphasized that this schematic has been derived from a systematic and comprehensive review of the literature included in this SLR. For this purpose, we considered multiple factors, such as the variety of techniques and evaluation models proposed for each sentiment analysis type, as well as the comparative performance of different methods against baseline results.

Now, we provide a detailed synthesis for each category of transformer-based sentiment analysis proposed in Sect. 4.3, Contextual-Adaptive, LLMs, Lightweight, and Hybrid models:

5.2.1 Contextual-adaptive models

Contextual-adaptive models represent the dominant paradigm in recent sentiment analysis research on social media text. These models leverage the powerful contextual understanding of pretrained transformers (primarily BERT, RoBERTa, and their variants) and then adapt or extend them—through domain-specific pretraining, fine-tuning, ensemble, hybrid architectures, auxiliary sentence techniques, weak supervision, or multimodal fusion—to address the unique challenges posed by real-world data.

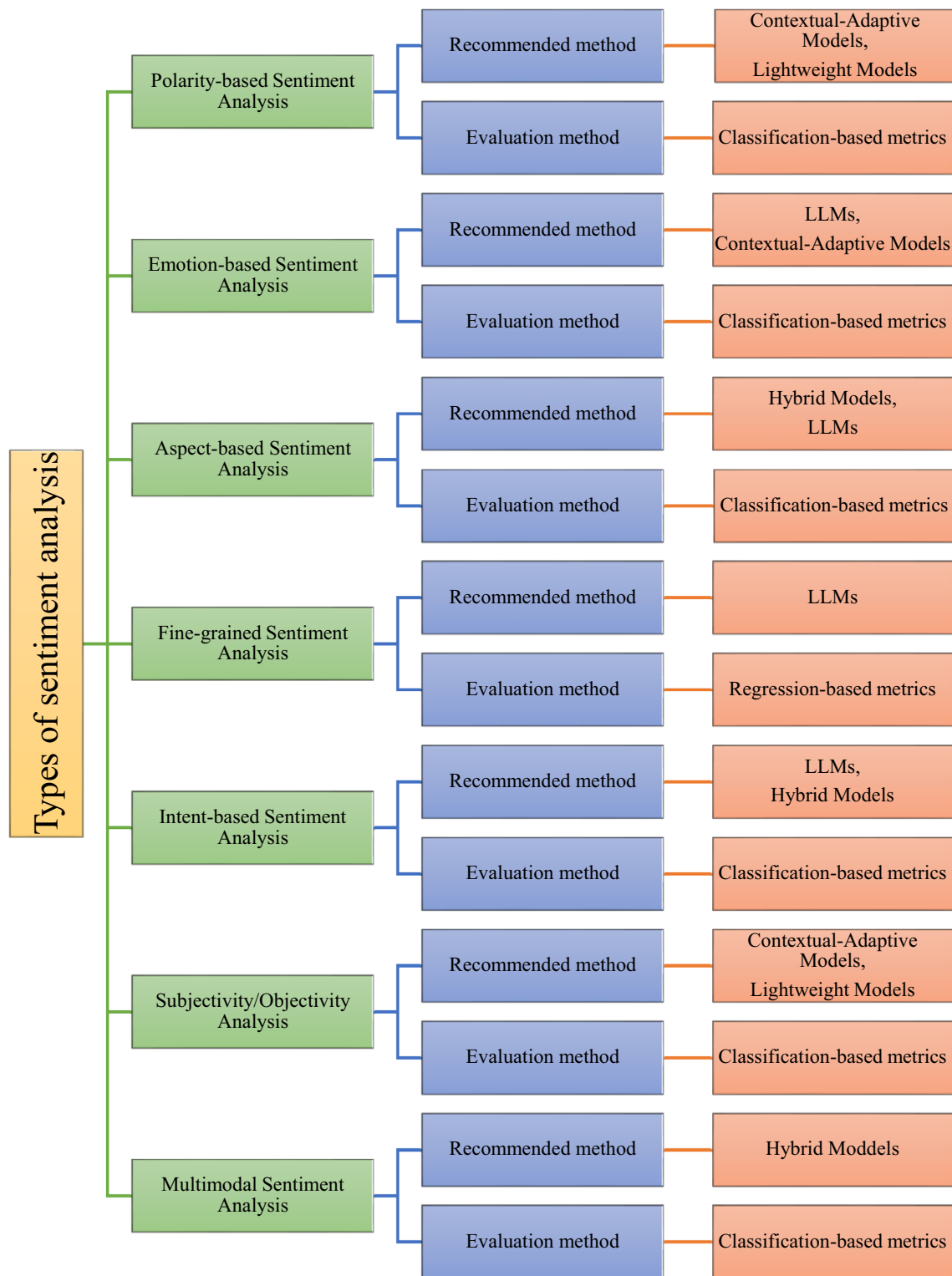


Fig. 12 Schematic of development guidelines for transformer-based sentiment analysis research

Social media text is typically short, noisy, informal, sarcastic, code-mixed, implicit, domain-specific, or expressed in low-resource languages. Traditional approaches (rule-based systems, classical machine learning such as SVM, Naïve Bayes, or Logistic Regression, and older topic models like LDA) consistently fail to handle these complexities effectively. In response, contextual-adaptive models are specifically tailored—either by continued pretraining on domain-relevant corpora (e.g., Twitter, financial news, or medical texts) or by designing purpose-built architectures—to align closely with the linguistic patterns and requirements of the target domain.

Across studies, these transformer-based approaches uniformly outperform traditional baselines, often by substantial margins, confirming their status as the new standard. By combining general-purpose contextual knowledge with targeted adaptation strategies, contextual-adaptive models achieve significantly higher accuracy and robustness in sentiment analysis tasks on challenging social media datasets.

Regarding the development guidelines for transformer-based sentiment analysis presented in Fig. 12, several recommended use cases for contextual-adaptive models can be identified, such as:

- **Polarity-Based Sentiment Analysis:** Contextual-adaptive models are exceptionally well-suited for polarity classification because they understand words in context rather than in isolation. These models handle negation, double negatives, and sentiment shifters with remarkable accuracy.
- **Emotion-Based Sentiment Analysis:** Detecting specific emotions requires understanding subtle linguistic cues, figurative language, cultural references, and implicit emotional signals. Contextual-adaptive models excel in this area because they can recognize implicit emotional expressions, sarcasm and irony, intensity gradations, and mixed emotions.
- **Subjectivity/Objectivity Analysis:** Contextual-adaptive models effectively distinguish between subjective and objective statements by understanding the semantic role of words in context. They recognize hedging language, evaluative adjectives, comparative constructions, and the difference between reported speech and personal assertion.

5.2.2 Large language models (LLMs)

Large language models (LLMs), built on decoder-only or generative transformer architectures with billions of parameters (e.g., GPT-3.4, LLaMA-2, PaLM-540B, Falcon, Mistral), represent a major shift away from the encoder-only paradigm in the previous section. Thanks to their massive scale and pretraining on diverse internet text, these models

can perform complex sentiment-related tasks—ranging from standard polarity classification to aspect-sentiment triplet extraction, intent detection, and bias analysis—using primarily in-context learning, few-shot examples, or carefully designed prompts, often with little or no task-specific labeled data or traditional fine-tuning.

The core promise explored in these studies is the potential to drastically reduce or eliminate the need for large annotated datasets and extensive fine-tuning. Instead of adapting a moderately sized encoder model to a specific domain, researchers investigate whether a single, general-purpose LLM can be steered toward high performance simply through prompt engineering, chain-of-thought reasoning, or weak-supervision pipelines that distill LLM judgments into smaller deployable models.

Despite encouraging results in controlled or clean-text settings, a consistent pattern emerges: performance remains highly brittle and heavily dependent on prompt design (choice of role, output format, few-shot examples, instructive vs. contextual framing). Moreover, when applied to the noisy, informal, sarcastic, or short social media texts that BERT-family models were explicitly adapted to handle, most LLMs exhibit significant performance degradation.

The development guidelines shown in Fig. 12 indicate several use cases where LLMs are recommended, including:

- **Emotion-Based Sentiment Analysis (Advanced):** Large language models demonstrate superior performance on advanced emotion detection tasks by leveraging world knowledge and common-sense reasoning, cultural and contextual understanding, tracking complex emotional narratives across longer texts, and identifying implicit and subtle emotions through sophisticated pragmatic knowledge.
- **Aspect-Based Sentiment Analysis:** LLMs are particularly effective at aspect-based sentiment analysis because of their zero-shot and few-shot capabilities, which reduce the need for extensive labeled data; their ability to discover open-ended aspects not predefined in advance; their detection of implicit aspects; and their natural handling of complex, hierarchical, or overlapping aspect relationships.
- **Fine-Grained Sentiment Analysis:** Fine-grained sentiment analysis can be excelled using large language models by distinguishing subtle intensity levels through understanding of intensifiers and downtoners, nuanced lexical choices, punctuation and emphasis cues, and comparative or superlative constructions.
- **Intent-Based Sentiment Analysis:** Large language models perform strongly in intent-based sentiment analysis due to their grasp of pragmatics, including interpretation of speech acts, recognition of implicit versus explicit intents, and handling of multi-intent utterances. Common intent categories they can identify include complaints, inquiries,

suggestions, praise, requests, comparisons, and churn signals.

5.2.3 Lightweight models

Lightweight models represent a pragmatic response to the real-world deployment barriers posed by full-sized transformers. Rather than pursuing ever-higher accuracy through larger models, these approaches prioritize speed, reduced memory footprint, lower inference latency, and feasibility on resource-constrained devices (edge hardware, mobile phones, low-cost servers, or high-throughput production systems). To achieve this, researchers employ classic compression techniques—knowledge distillation, structured pruning of attention heads, and direct adoption of distilled architectures such as DistilBERT—while accepting a deliberate trade-off: slightly or moderately lower performance in exchange for dramatic gains in efficiency.

Across these studies, DistilBERT emerges as the dominant choice, consistently outperforming traditional machine learning baselines (SVM, Naïve Bayes, RNNs) and older deep learning models, yet remaining smaller and faster than BERT-base. Despite these advances, recurring limitations reveal that the efficiency–accuracy balance is still fragile: evaluations are typically confined to one or a few datasets, raising serious concerns about generalizability across domains, languages, or noise levels; aggressive sequence truncation risks discarding critical context in longer texts; and the compression or distillation processes themselves often remain computationally expensive. Consequently, while lightweight models have clearly made sentiment analysis far more deployable, achieving truly robust performance on noisy, diverse, real-world data with tiny models remains an open challenge.

Based on Fig. 12, the following use cases are recommended for lightweight models:

- **Polarity-Based Sentiment Analysis (Production Scale):** Lightweight models excel in production-scale polarity-based sentiment analysis due to their high throughput, enabling processing of thousands of texts per second; low latency for real-time responses in user-facing applications; suitability for edge deployment on mobile apps, IoT devices, and browser-based applications; and overall cost efficiency through reduced reliance on expensive GPU instances.
- **Subjectivity/Objectivity Analysis (Preprocessing Filter):** Lightweight models are well-suited for subjectivity/objectivity analysis as a preprocessing filter because this is typically a simpler binary classification task where they retain strong accuracy; there is higher error tolerance since minor misclassifications do not significantly

impact downstream processes; and they enhance pipeline efficiency by providing the fastest processing in the initial stage, allowing more resource-intensive models to be used in later precision-critical stages.

5.2.4 Hybrid models

These studies represent the "enhancement and hybridization" wave of transformer-based sentiment analysis. Rather than accepting vanilla BERT or RoBERTa as the final answer, they all seek to push performance higher on difficult tasks by augmenting, extending, or combining BERT-family models with additional mechanisms. The shared motivation is clear: plain fine-tuned BERT is strong but still insufficient for the hardest real-world sentiment scenarios, so extra architectural or learning innovations are required to close the remaining accuracy gap.

Common additions include: CNN, TextCNN, BiLSTM, BiGRU, CABiLSTM, Graph Neural Networks—GCN, GAT, hypergraph, External sentiment lexicons, Multimodal fusion modules, Contrastive or self-supervised auxiliary losses, Stacking, ensemble, meta-learners, LightGBM on top of BERT variants, heterogeneous stacking. They are targeted at "harder" sentiment problems, but they have High computational complexity, memory demands, long training time, and Heavy reliance on quality of external components (lexicons, dependency parsers, prompts, graphs).

This large group of studies collectively shows that, while plain fine-tuned BERT-family models have become the uncontested baseline, researchers hybridize them with CNNs, RNNs, graphs, lexicons, contrastive objectives, or multimodal fusion modules to squeeze out the last few percentage points on challenging sentiment tasks.

Several recommended use cases for hybrid models are:

- **Aspect-Based Sentiment Analysis (Structured Extraction):** Hybrid models excel in aspect-based sentiment analysis through structured extraction by combining contextual representations from transformer encoders, graph neural networks to leverage syntactic structures and propagate sentiment signals, and attention mechanisms to align aspects with their corresponding opinion expressions, even across distant parts of the sentence.
- **Intent-Based Sentiment Analysis (Enterprise Applications):** Hybrid models are particularly effective for enterprise intent-based sentiment analysis due to the need for consistent classification into predefined intent taxonomies, reliable confidence scores for escalation, interpretability for regulated environments, and low-latency processing at scale. They achieve this by integrating transformer encoders for contextual understanding, sequential layers for modeling conversation flow, external knowledge

sources such as customer history and business rules, and multi-task learning to jointly optimize intent and sentiment prediction.

- **Multimodal Sentiment Analysis:** Hybrid models perform strongly when text alone is insufficient for accurate interpretation. They excel by incorporating cross-modal attention mechanisms to allow one modality to inform another, fusion modules that combine representations from different modalities at early, late, or hybrid levels, and alignment mechanisms to maintain temporal synchronization in dynamic inputs such as video.

We conclude this section by summarizing the discussions related to the five research questions presented in Table 2.

RQ1. How has sentiment analysis evolved with the advent of transformers?

As outlined in Sect. 4.1, sentiment analysis initially relied on lexicon-based techniques and traditional machine learning algorithms. The adoption of deep learning methods enhanced text understanding, yet limitations persisted in capturing long-term dependencies and contextual nuances. Transformer models, through the use of self-attention mechanisms and transfer learning, have substantially advanced the field by enabling more precise representations of meaning, structure, and linguistic subtleties, thereby significantly enhancing sentiment analysis performance.

RQ2. Which transformer-based models have been employed for sentiment analysis?

According to Sect. 4.3, transformer-based approaches can be grouped into four categories: contextual-adaptive models, large language models, lightweight models, and hybrid models. Among these, contextual-adaptive models emerge as the most widely adopted, as illustrated in Fig. 8.

RQ3. What are the key datasets used for evaluating transformer-based sentiment analysis models?

To address the third research question, this review identified the principal datasets used across the analyzed studies, together with their defining characteristics. Figure 6 provides a comparative assessment of their relevance within the sentiment analysis domain, underscoring their role in benchmarking model performance.

RQ4. Which metrics are used to evaluate transformer-based sentiment analysis models?

As discussed in Sect. 4.4, the reviewed studies employed a variety of evaluation measures. These can broadly be categorized into two groups: classification-oriented metrics and regression-oriented metrics. This categorization highlights the dual perspectives from which model effectiveness is assessed.

RQ5. How has the maturity and scientific quality of transformer-based sentiment analysis research evolved in recent years?

Section 4.5 examined four quality assessment factors to track progress in sentiment analysis research. The findings, summarized in Fig. 11 and Table 10, reveal a clear trajectory of advancement in recent years, with LLM-based approaches playing a pivotal role in enhancing both the maturity and the scholarly quality of sentiment analysis studies.

6 Threats to validity

One of the main threats to the validity of this systematic literature review is incompleteness. The risk of this threat highly depends on the selected list of keywords forming search queries. To decrease the risk of an incomplete keyword list, we have used an iterative approach to keyword-list construction. We constructed five queries, each for one of our research questions. To ensure that we include as many articles as possible that are in line with the goals of this article. As in any other literature survey, we are also prone to a publication bias. To mitigate this threat, we complemented a digital library search with other strategies. We reviewed the articles to ensure that only papers from high-level conferences were included in this study. To ensure the quality and reliability of our study, we defined a systematic procedure for conducting our survey, including research questions to answer and searching strategies. Consequently, the selection of primary studies followed a strict protocol of PRISMA. The guidelines were initially introduced by Moher et al. [85], and an updated version, known as PRISMA 2020, was published in 2021 [86]. Another threat to validity we would like to highlight is our subjectivity in screening, data extraction, and classification of the studied papers. To mitigate this threat, each step was initially performed by one author (the first author of this paper) and subsequently cross-checked by a second author. The inter-rater agreement achieved across all phases was very high. Then, an external evaluator (the leading/supervising author) verified the agreement between the first and the second authors. A similar threat concerns whether our taxonomies are reliable enough for analysing and classifying extracted data. To mitigate this threat, we used an iterative content analysis method to continuously develop each taxonomy. New concepts that emerged when studying the papers were introduced into a taxonomy, and changes were made accordingly. These taxonomies were discussed among all the authors and agreed upon in their final form.

7 Conclusion

Sentiment analysis has emerged as a pivotal tool for understanding public opinion, enhancing services, supporting data-driven decision-making, and managing crises. Consequently, the development of accurate sentiment analysis

approaches, particularly those leveraging transformer-based architectures, has become a primary research priority in natural language processing (NLP).

This study conducts a systematic literature review (SLR) of 347 articles published between Jan 2020 and Aug 2025 to trace the evolution of transformer-driven sentiment analysis. Five research questions were formulated to investigate key dimensions of the field, including methodological advances, state-of-the-art models, benchmark datasets, evaluation metrics, and the growing influence of large language models (LLMs). These questions aim to map progress, categorize tools, and highlight both challenges and opportunities for future work.

The main findings of this SLR can be summarized as follows:

- **Applications:** Sentiment analysis has been applied across a wide range of domains, including marketing (e.g., product and service reviews), social media analytics (e.g., Twitter and other platforms), public health (e.g., attitudes toward vaccines), finance (e.g., volatility forecasting), entertainment (e.g., film reviews), and political discourse analysis. Each application domain presents unique challenges that require tailored methodological approaches.
- **Methods and models:** Transformer architectures such as BERT, RoBERTa, and more recent LLMs have substantially advanced both the accuracy and contextual depth of sentiment analysis through attention mechanisms and large-scale pretraining. Lightweight models (e.g., DistilBERT), hybrid architectures, and lexical-symbolic approaches have also proven effective in demanding tasks such as aspect-based sentiment analysis (ABSA) and mul-

timodal sentiment analysis.

- **Evaluation metrics:** Performance has been primarily assessed using measures such as accuracy, F1-score, recall, AUC-ROC, and regression metrics such as MAE and RMSE. Using a combination of these measures enables more comprehensive and robust evaluation, particularly in multilingual and multimodal contexts.
- **Datasets:** Widely used benchmark datasets include SemEval, CMU-MOSEI, CMU-MOSI, IMDB, and Sentiment140. These resources, however, present challenges such as class imbalance, scarcity of data for low-resource languages, and multimodal complexity, underscoring the need for more diverse and balanced datasets.
- **Quality of studies:** The evaluation of included works indicates both quantitative and qualitative growth in transformer-based sentiment analysis research from Jan 2020 to Aug 2025, demonstrating the dynamism of the field and its potential for interdisciplinary and innovative applications.

Based on the gaps identified, this study proposes a roadmap for future research. Key directions include advancing methods to capture linguistic subtleties and implicit emotions, addressing unbalanced and noisy data, optimizing models for resource-constrained and real-time scenarios, and designing multimodal, multilingual, and interpretable systems, with continuous attention to the ethical dimensions of sentiment analysis.

Appendix

Publication venue	Type	Number	Impact factor
Expert Systems with Applications	Journal Article	11	7.5
IEEE Access	Journal Article	9	3.6
Information Processing & Management	Journal Article	5	6.9
Knowledge-Based Systems	Journal Article	5	7.6
Applied Soft Computing	Journal Article	5	6.6
The Journal of Supercomputing	Journal Article	4	2.7
Multimedia Tools and Applications	Journal Article	3	-
Procedia Computer Science	Journal Article	3	-
Information Sciences	Journal Article	3	6.8
Soft Computing	Journal Article	3	2.5
Complex & Intelligent Systems	Journal Article	3	4.6
IEEE Transactions on Affective Computing	Journal Article	3	9.8
Results in Engineering	Journal Article	3	7.9
SN Computer Science	Journal Article	3	-
Applied Intelligence	Journal Article	2	3.5
IEEE Transactions on Multimedia	Journal Article	2	9.7
International Journal of Data Science and Analytics	Journal Article	2	2.8
Neurocomputing	Journal Article	2	6.5
Natural Language Processing Journal	Journal Article	2	-
Neural Computing & Applications	Journal Article	2	-
Future Generation Computer Systems	Journal Article	2	6.1
Cognitive Computation	Journal Article	2	4.3
New Generation Computing	Journal Article	2	2.8
The Journal of Intelligent Information Systems	Journal Article	2	3.4
Neural Computing and Applications	Journal Article	2	-
Artificial Intelligence Review	Journal Article	1	13.9
Chaos, Solitons & Fractals	Journal Article	1	5.6
Computational Intelligence	Journal Article	1	7.4
Computer Speech & Language	Journal Article	1	3.4
Computers and Electrical Engineering	Journal Article	1	4.9
IEEE Signal Processing Letters	Journal Article	1	3.9
IEEE Transactions on Computational Social Systems	Journal Article	1	4.9
IEEE Transactions on Emerging Topics in Computational Intelligence	Journal Article	1	6.5
IEEE Transactions on Games	Journal Article	1	2.8
Information and Software Technology	Journal Article	1	4.3
Information Fusion	Journal Article	1	15.5
International Journal of System Assurance Engineering and Management	Journal Article	1	1.4
Journal of Computational Social Science	Journal Article	1	2.3
Journal of Computer Science and Technology	Journal Article	1	1.3
Journal of Transport Geography	Journal Article	1	6.3
Knowledge and Information Systems	Journal Article	1	3.1
Language Resources and Evaluation	Journal Article	1	1.8

Publication venue	Type	Number	Impact factor
Multimedia Systems	Journal Article	1	3.1
Neural Networks	Journal Article	1	6.3
Pattern Analysis and Applications	Journal Article	1	2
TechTrends	Journal Article	1	3.8
Customer Needs and Solutions	Journal Article	1	–
Data Science for Transportation	Journal Article	1	–
International Journal of Information Technology	Journal Article	1	–
Journal of Open Innovation: Technology, Market, and Complexity	Journal Article	1	–
Journal of the Knowledge Economy	Journal Article	1	–
Journal of Biomedical Informatics	Journal Article	1	4.5
Pattern Recognition and Computer Vision	Book	2	–
Social Networks Analysis and Mining	Book	1	–
Engineering Applications of Artificial Intelligence	Book	1	–
2023 Companion Proceedings of the ACM Web Conference	Conference Proceedings	2	–
2021 Asian Conference on Innovation in Technology (ASIANCON)	Conference Proceedings	1	–
2022 2nd International Conference on Digital Futures and Transformative Technologies (ICoDT2)	Conference Proceedings	1	–
2022 9th International Conference on Computing for Sustainable Global Development (INDIACom)	Conference Proceedings	1	–
2022 IEEE International Conference on Data Mining Workshops (ICDMW)	Conference Proceedings	1	–
2023 9th International Conference on Signal Processing and Communication (ICSC)	Conference Proceedings	1	–
2023 9th International Conference on Web Research (ICWR)	Conference Proceedings	1	–
2023 International Conference on Intelligent Systems for Communication, IoT and Security (ICISCoIS)	Conference Proceedings	1	–
2020 25th International Conference on Pattern Recognition (ICPR)	Conference Proceedings	1	–
2020 IEEE 5th International Conference on Computing Communication and Automation (ICCCA)	Conference Proceedings	1	–
2020 International Conference on Communications, Information System and Computer Engineering (CISCE)	Conference Proceedings	1	–
2020 Natural Language Processing and Chinese Computing 9th CCF International Conference	Conference Proceedings	1	–
2021 11th International Conference on Cloud Computing, Data Science & Engineering (Confluence)	Conference Proceedings	1	–
2021 6th International Conference on Computer Science and Engineering (UBMK)	Conference Proceedings	1	–
2021 ACM Southeast Conference	Conference Proceedings	1	–
2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC)	Conference Proceedings	1	–

Publication venue	Type	Number	Impact factor
2021 Proceedings of the 29th ACM International Conference on Multimedia	Conference Proceedings	1	–
Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing	Conference Proceedings	1	–
2022 15th International Conference on Computational Processing of the Portuguese Language	Conference Proceedings	1	–
2022 5th International Conference on Artificial Intelligence and Pattern Recognition	Conference Proceedings	1	–
2022 IEEE International Conference on Data Mining Workshops (ICDMW)	Conference Proceedings	1	–
2022 IEEE International Conference on Multimedia and Expo (ICME)	Conference Proceedings	1	–
2022 IEEE 10th International Conference on Information, Communication and Networks (ICICN)	Conference Proceedings	1	–
2022 IEEE International Conference on Big Data (Big Data)	Conference Proceedings	1	–
2022 IEEE International Power and Renewable Energy Conference (IPRECON)	Conference Proceedings	1	–
2023 Mexican International Conference on Artificial Intelligence(MICAI)	Conference Proceedings	1	–
2023 12th International Conference on Computing and Pattern Recognition	Conference Proceedings	1	–
2023 15th International Conference on COMMunication Systems & NETWORKS (COMSNETS)	Conference Proceedings	1	–
2023 26th International Conference on Computer and Information Technology (ICCIT)	Conference Proceedings	1	–
2023 31st ACM International Conference on Multimedia	Conference Proceedings	1	–
2023 3rd International Conference on Artificial Intelligence(CAAI)	Conference Proceedings	1	–
2023 3rd International Conference on Intelligent Technologies (CONIT)	Conference Proceedings	1	–
2023 4th ACM International Conference on AI in Finance	Conference Proceedings	1	–
2023 4th International Conference on Machine Learning and Computer Application	Conference Proceedings	1	–
2023 Fourth International Conference on Intelligent Data Science Technologies and Applications (IDSTA)	Conference Proceedings	1	–
2023 IEEE 11th Conference on Systems, Process & Control (ICSPC)	Conference Proceedings	1	–
2023 IEEE 5th International Conference on Power, Intelligent Computing and Systems (ICPICS)	Conference Proceedings	1	–
2023 IEEE 8th International Conference for Convergence in Technology (I2CT)	Conference Proceedings	1	–
2023 IEEE Asia–Pacific Conference on Computer Science and Data Engineering (CSDE)	Conference Proceedings	1	–
2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)	Conference Proceedings	1	–
2023 IEEE Latin American Conference on Computational Intelligence (LA–CCI)	Conference Proceedings	1	–
2023 International Conference on Data Science and Its Applications (ICoDSA)	Conference Proceedings	1	–

Publication venue	Type	Number	Impact factor
2023 International Conference on Machine Learning and Applications (ICMLA)	Conference Proceedings	1	–
2023 International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)	Conference Proceedings	1	–
2024 International Conference on Information Technology (InCITE)	Conference Proceedings	1	–
2024 16th International Conference on Advances in Social Networks Analysis and Mining (ASONAM)	Conference Proceedings	1	–
2024 20th CSI International Symposium on Artificial Intelligence and Signal Processing (AISP)	Conference Proceedings	1	–
2024 23th International Conference on Artificial Intelligence and Soft Computing(ICAISC)	Conference Proceedings	1	–
2024 24th International Conference on Web Engineering(ICWE)	Conference Proceedings	1	–
2024 37th Australasian Joint Conference on Artificial Intelligence(AI)	Conference Proceedings	1	–
2024 7th Chinese Conference on Pattern Recognition and Computer Vision (PRCV)	Conference Proceedings	1	–
2024 World Conference on Information Systems and Technologies(WorldCIST)	Conference Proceedings	1	–

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Declarations

Competing interests The authors declare no competing interests.

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